The dynamics of ‘don’t know’



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Abstract

This report investigates how uncertainty affects the robustness of a decentralised decision-making algorithm called the adaptive voter model. We extend the n state adaptive voter model to include uncertain belief states, a random state change event and the presence of agents that do not change their opinion. We analytically demonstrate how the addition of an uncertain truth state improves the fault tolerance and the noise tolerance of a three-state adaptive voter model in certain parameter regimes. In addition, we establish good agreement between the analytical results and numerical results obtained from agent-based simulations.

Author's declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Research Degree Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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1 Introduction

1.1 Motivation

Collective decision-making is a term used to describe the process in which a group of agents attempt to reach a shared decision without centralised control. There are numerous examples of this phenomenon occurring in the natural world. A notable and often cited example is the ability of honeybees and ants to choose a suitable nest site using collective and decentralised information processing [1]. There is an extensive body of literature concerning collective decision-making from a variety of different fields such as ethology, control theory, robotics and statistical physics [2].

For example, the field of swarm robotics seeks to develop robotic systems that utilise this phenomenon of emergent collective intelligence to make decisions without top-down control or global information about the system. An ideal robot swarm will be scalable to different swarm sizes, tolerant to individual failure and robust to different types of noise [2]. Swarm robotic systems are particularly suited for applications that require spatial covering and operation in dangerous environments [3]. Possible applications include search and rescue missions, cleaning pollutants from large areas, monitoring an environment for hazardous events, medicine and swarm robotic inspection systems [3].

Despite its many potential advantages, swarm robotics has yet to be widely adopted as a method for solving real-world problems due to a number of limitations. One of these limitations is the robustness of a robotic swarm to noise that equally affects all members of the swarm such as processing limitations or sensing errors [4]. We shall refer to this form of robustness as noise tolerance. Another important limitation is the robustness of a robotic swarm to individual failure of agents which we shall refer to as fault tolerance [4, 5]. A proposed method of improving the fault tolerance of robotic swarms is to endow robots with the ability to detect faults in their neighbours [4, 5]. This approach is referred to as exogenous fault detection in the literature. The efficacy of this method relies on the accuracy of the individual robot's fault detection system. In order to avoid this somewhat circular reasoning we want to explore the possibility of building fault tolerance into the design of decentralised decision-making algorithms following the example of Crosscombe et al., [4]

The aim of this project is to investigate whether the noise tolerance, robustness to noise, and the fault tolerance, robustness to failure, of an existing decentralised decision-making algorithm can be improved through the inclusion of explicit uncertainty. We believe that the incorporation of uncertainty can help to avoid polarisation of beliefs and potentially decrease the convergence time to a shared belief, also referred to as consensus time, by allowing agents with opposing beliefs to moderate each other. We hold that the development of more robust decentralised decision-making algorithms is a necessary prerequisite for wide scale adoption of swarm robotics as a tool for solving complex real-world challenges. Furthermore, we hope that a greater understanding of robust decentralised decision-making may find a broader application in areas such as the study of opinion formation on social networks.

1.2 Related work

The problem of achieving an optimal collective decision in a group of agents is known as a discrete consensus achievement problem [2]. Discrete consensus achievement problems have been studied in variety of contexts. Notable examples include the study of decision-making in teams of agents by the artificial intelligence community [6] and decentralised object classification using sensors by the robotics community [7]. Some of the decision-making approaches rely on complex or centralised communication strategies that do not scale well to large group sizes. In contrast, the statistical physics community has proposed a range of models to describe the opinion dynamics of groups of interacting agents [8] that do not suffer the same scaling problems.

The discrete consensus achievement problem can be modelled using an abstraction called the best-of-n problem [2]. This requires a multi-agent swarm to collectively choose the best proposition out of n mutually exclusive propositions. Each proposition is characterized by a quality and the collective decision must be made with only local feedback and learning. Each agent has a belief state which represents their opinion about which proposition is best and the agents change their belief state by applying a given decision rule [9].

One such set of decision rules is the voter model, a paradigm model in the statistical physics community that has been applied to the best-of-n problem, amongst other things. The model consists of a a network of N nodes, which represent agents, connected by L undirectional links, which represent contacts between agents. Each node i is associated with a binary variable which represents their opinion about a single proposition. At each time step two nodes i and j, where j is a neighbour of i, are selected uniformly at random and node i adopts the opinion of node j [8].

There have been numerous modifications to the voter model. We shall review those that are most pertinent to the current work. Starini et al., [10] propose a multi-state voter model in which each agent can be in one of N different exclusive states. This is an important development in the context of the best-of-n problem as there are n mutually exclusive propositions to choose from however this model is lacking any explicit notion of uncertainty.

The AB-model [11] extends the voter model by including a ternary variable to represent an agent's opinion about a proposition. In this model an agent has to pass through an intermediate state AB in order to move from state A to state B. Patriarca et al., [11] found that the addition of the intermediate state did not change the system dynamics significantly. In a similar vein, Perron et al., [12] investigated the effect of introducing an intermediate state for binary consensus on complete graphs. Perron et al., [12] explore the binary consensus problem where each node in the network initially observes a binary variable and then has to decide which of the two possible states was initially held by the majority. The intermediate truth state 'e' represents an agent being uncertain of the majority state [12]. It is shown that the introduction of this extra state decreases the time taken to reach the correct consensus on the complete graph. Furthermore, it is shown that the addition of uncertainty increases the reliability of reaching the correct consensus on the complete graph. These approaches introduce a notion of uncertainty but are lacking an investigation of the n proposition case.

The weighted voter model incorporates feedback on the quality of agent beliefs. In this

model each proposition is characterised by a given quality p. Within the framework of the best-of-n problem, a solution to the weighted voter model is successful if consensus is reached on the proposition with the highest quality [9]. The weighted voter model is analysed in the two proposition case, $n = 2$, of the best-of-n problem by Valentini et al., [9]. It is shown that the weighted voter model exhibits better network size scaling, in terms of time take to reach consensus and decision accuracy, than the original voter model. Furthermore, it is shown that this model is more robust to noise. This approach introduces a notion of quality in the characterisation of the propositions however it only investigates the two proposition case and is lacking a representation of uncertainty.

The weighted voter model has been extended by Crosscombe et al., [4] to incorporate a ternary variable to represent an agent's opinion about a single proposition [4]. This ternary variable takes a value from the set $\{0, \frac{1}{2}, 1\}$ where $\frac{1}{2}$ is an intermediate truth state which represents an 'uncertain' or 'borderline' opinion [4]. Crosscombe et al., [4] demonstrate that this extended model exhibits a trade-off between speed of convergence to a shared belief and fault tolerance, which is measured as robustness to the effect of malfunctioning agents. Specifically, they show that the extended model has a higher fault tolerance but displays a slower speed of convergence to a shared belief than the traditional weighted voter model. Despite presenting preliminary results on the $n > 2$ proposition case, this work is lacking a thorough investigation of the general n proposition case. Furthermore it does not investigate how the addition of uncertainty affects the noise tolerance of the models.

One development of the voter model that is especially relevant to the current work is the introduction of 'zealots' by Mobilia et al., [13], where a zealot is an agent that never changes its opinion. Mobilia et al., [13] found that the addition of only a few zealots prevents convergence on a consensus opinion and even prevents the formation of a large majority. This study is important for the current work as the notion of zealots provides a convenient modelling tool for exploring fault tolerance.

Another apposite development, that is referred to as the noisy voter model, is the introduction of random changes of state. Carro et al., [14] show that when the noise parameter exceeds a critical value, a finite size system will transition from a consensus state, where all agents converge to the same opinion, to a polarised state, where the two opposing opinions survive. This demonstrates that the voter model has a low noise tolerance. This is relevant to our problem as the introduction of random changes of state provides a convenient modelling tool for exploring noise tolerance.

All of the developments reviewed so far consider a static, connected interaction network. The work of Mobilia et al., [13] demonstrates that convergence to a consensus state in the voter model on a static, connected network is impossible in the presence of zealots. In an attempt to develop a decentralised decision-making algorithm with a higher fault tolerance we will instead, consider a dynamic interaction network.

The role of network topology has been further explored by investigating a voter model with an interaction network which co-evolves with the agent dynamics [15]. This type of model is called the adaptive voter model. It introduces the notion of homophily, the tendency of individuals to bond with similar others, into the voter model. In addition to the opinion adoption

event present in the original voter model, the adaptive voter model introduces a link rewiring event. The link rewiring event involves one agent breaking its link to another agent and then forming a new link to a different agent that holds the same opinion [16].

The adaptive voter model has been shown by Vazquez et al., [17] to display an absorbing transition from a connected to a fragmented network at a critical value which depends on the average degree of the network. This fragmentation transition can be computed analytically by a motif expansion proposed by Bohme and Gross [18] and a 'triple-jump' method proposed by Silk et al., [16]. The adaptive voter model has been extended to include an arbitrary number of propositions by Bohme and Gross [19]. Bohme and Gross [19] present detailed analytical results on the fragmentation transition in a three-state model. These approaches are missing an explicit representation of uncertainty and they do not consider the fault or noise tolerance of the models.

In addition to exploring the fragmentation transition, a general analytical method has been proposed for computing the consensus time in the adaptive voter model by Rogers and Gross [20]. It is shown that the scaling of the time taken to converge on a consensus state as a function of the number of nodes in the network is greater than that for the standard voter model. This approach only considers the $n = 2$ proposition case and does not investigate fault tolerance or noise tolerance.

1.3 Project aims and objectives

Given the work of Mobilia et al,. [13] we suggest that a decentralised decision-making algorithm with a dynamic interaction network will have a higher fault tolerance that one with a static interaction network. We will therefore investigate the adaptive voter model as a suitable decentralised decision making-algorithm for solving the best-of-n problem. Following the work of Crosscombe et al., [4], we propose that the fault tolerance and noise tolerance of the adaptive voter model may be improved by explicitly including the possibility of uncertainty in agents' belief states.

The aims of this project are threefold. We will provide a modelling framework for an extended adaptive voter model that includes uncertain beliefs, weighted propositions, some notion of inconsistency between belief states, a random state change event and the presence of zealots. We will use this modelling framework to explore the effect of uncertainty on the fault tolerance and noise tolerance of the model with agent-based simulations. In addition, we will extend the analytical methods for calculating the fragmentation threshold and the consensus time in the three-state adaptive voter model to a three-state adaptive voter model that includes an uncertain belief state. We will compare the results from this analysis with results on the standard adaptive voter model in the literature, to determine the effect of uncertainty on the fault tolerance and noise tolerance of the adaptive voter model.

1.4 Report plan

In order to achieve the aims stated above, we will begin by constructing a general modelling framework in Chapter 2. We shall then proceed, in Chapter 3, to analytically calculate the fragmentation threshold and the consensus time of the three-state adaptive voter model with

uncertainty using motif expansion methods employed by Bohme and Gross [18, 19]. We shall then use the developed analytical framework to compare the fault and noise tolerance of this model with the three-state adaptive voter model without uncertainty. The effect of uncertainty on the fault and noise tolerance of the n state extended adaptive voter model will be examined through agent-based simulation in Chapter 4. Finally, we conclude with a summary and discussion about the significance of the results in an academic and an applied context.

2 Modelling framework

As described in Chapter 1, the problem of achieving an optimal collective decision in a group of agents, with only decentralised algorithms, is known as a discrete consensus achievement problem. The best-of-n problem is an abstraction that provides a modelling framework for this type of problem [9].

The best-of-n problem requires a group of agents to collectively choose the best proposition out of n mutually exclusive propositions. Each of these n propositions is characterised by a quality. Each agent has a belief state which represents their opinion about which proposition they think is best. The agents change their belief state by applying a given decision rule. The collective decision must be made with only local feedback and learning [9].

We extend the adaptive voter model to the n proposition case and to include weighted propositions, uncertain belief states, a random state change event and the presence of zealots. We call this the ternary weighted adaptive voter model and we shall refer to it using the acronym T.V. . We also consider a special case of this model in which there are no uncertain belief state. We call this the binary weighted adaptive voter model and we shall refer to it using the acronym B.V. . The general modelling framework outlined above can be tailored to describe our extended models by describing exactly how propositions, belief states and decision rules will be captured. We will first give a complete description of how propositions, belief states and decision rules will be modelled for the T.V. model. We then discuss the simplifications required to adapt this description to the B.V. model.

2.1 Modelling propositions

As in the best-of-n problem, we shall define propositions to be mutually exclusive. We assume that in a real-world application agents would not have perfect knowledge about the quality of propositions. Given this assumption we model the measured quality, q_i , of a proposition , p_i , as a continuous random variable sampled from a Gaussian distribution with a given mean, μ_i, and variance, σ_i^2 ,

$$q_i \sim \mathcal{N}(\mu_i, \sigma_i^2). \tag{1}$$

2.2 Modelling belief states

In the best-of-n problem each agent has a belief state that expresses their opinion about each proposition. Each opinion is encoded as a binary variable $o_i \in \{0, 1\}$ with $o_i = 0$ representing that the agent believes that proposition p_i is not the best and $o_i = 1$ representing that the agent

believes that proposition p_i is the best. An agents belief state, s can then be encoded as an n-dimensional vector

$$s \in \{0,1\}^n : \sum_{i=1}^{n} o_i = 1, \tag{2}$$

where the condition $\sum_{i=1}^{n} o_i = 1$ enforces mutual exclusivity of the propositions. Using (2) we denote

$$s^i \equiv s : \quad o_i = 1 \wedge o_j = 0 \quad \forall j \in \mathbb{Z} : \quad j \in \{1,\ldots,n\} - \{i\} \tag{3}$$

as the belief state in which an agent believes that proposition p_i is the best. Following the work of Crosscombe and Lawry [21] we use three-valued logic to model an agent's opinion about a proposition. In our extended model an agent's opinion about a particular proposition is encoded as a ternary variable $o_i \in \{0, \frac{1}{2}, 1\}$. Opinions 0 and 1 are as above, whilst $\frac{1}{2}$ represents the opinion that agent j is uncertain whether proposition p_i is the best or not the best.

An agents belief state can then be encoded as an n-dimensional vector $t \in \{0, \frac{1}{2}, 1\}^n$ however for later work it is more convenient to represent an agents belief state in our extended model using set notation. In particular we can represent an agents belief state as

$$B \in \mathscr{P}(S) \setminus \emptyset, \tag{4}$$

where

$$S = \{s : s \in \{0,1\}^n : \sum_{i=1}^{n} o_i = 1\}. \tag{5}$$

Notice that any such B can be represented by a vector $t \in \{0, \frac{1}{2}, 1\}^n$ such that,

$$t_i = \begin{cases} 1 : \min\{s^i : s \in B\} = 1 \\ 0 : \max\{s^i : s \in B\} = 0 \\ \frac{1}{2} : \text{otherwise} \end{cases} \tag{6}$$

To clarify the equivalence between these two representations consider an example when $n = 3$. In the 3 proposition case a possible belief state in our extended model could be

$$t = \left(\tfrac{1}{2}, \quad \tfrac{1}{2}, \quad 0\right). \tag{7}$$

The equivalent set representation of this belief state would be

$$B = \left\{\left(1, \quad 0, \quad 0\right), \left(0, \quad 1, \quad 0\right)\right\} = \left\{s^1, s^2\right\}. \tag{8}$$

In the rest of this report we shall refer to a belief state when an agent is certain that one proposition is best

$$B = s^i : i \in \mathbb{Z} \wedge i \in \{1,\ldots,n\}, \tag{9}$$

as a certain belief state and we shall refer to a belief state when an agent is uncertain about which proposition is best

$$B \in \mathscr{P}(S) \setminus \emptyset \quad : \quad |B| \geq 2, \tag{10}$$

as an uncertain belief state. Given that each proposition has an associated quality and that a belief state is a set of opinions about each proposition we need to decide how to assign a quality to each belief state, Q_B. There are two distinct cases to consider: an agent has a certain belief state and an agent has an uncertain belief state. In the first case the quality of this belief state would just be the quality of the proposition that the agent has decided is the best

$$Q_B = q_i. \tag{11}$$

In the second case we choose to model the quality of the belief state as the arithmetic mean of the qualities of the propositions that the agent is yet to rule out as not the best. We suggest that this allows agents to gather more information about the quality of propositions before making a decision to rule propositions out. Given our set representation of belief states we can summarise these two cases into a single definition for the quality of a belief state as

$$Q_B = \overline{\left\{q_k : s^k \in B\right\}}. \tag{12}$$

2.3 Modelling agent interaction and decision rules

The T.V. model defines a network comprised of N nodes, representing agents, and L undirected links, representing contacts between agents. Each agent is associated with a set B that represents their belief state. We initialise the network as random graph, generated from the configuration model $G(n,p)$, with mean degree $\langle k \rangle = \frac{2L}{N}$ [22]. The initial belief state of each node is drawn uniformly at random from the set of all permissible belief states.

The time evolution of the network is determined through successive update steps. In each update step a link between nodes (α,β) is chosen from the set of all links, with a probability that is proportional to either the sum of, $(Q_{B_\alpha} + Q_{B_\beta})$, or the difference between, $(Q_{B_\alpha} - Q_{B_\beta})$ the qualities of the belief state of each node present in the link. The justification for the first of these choices is that in real-world systems it is often the most successful agents, where we measure success in terms of the payoff value of the agents belief state, whom have the most to gain by resolving conflicts with each other [21]. A possible justification for the second of these choices is based on the hypothesis that, in real-world systems, a better collective outcome is usually reached if conflict resolution between the least and the most successful agents is promoted. The link, (α,β), is considered to be inert if the belief states are equal, $B_\alpha = B_\beta$. If the link is inert then nothing happens and the network remains unchanged. If the belief states are different, $B_\alpha \neq B_\beta$, then this link is considered to be active and either a rewiring event, an opinion adoption event or a random state change event occurs on the link.

Considering first rewiring events, we propose that agents who have inconsistent belief states are more likely to disassociate from each other by breaking contact links than those with consistent belief states. Furthermore we suggest that agents who have broken a contact are more likely to associate with other agents who have a consistent belief state. Two agents are defined to have inconsistent belief states if the inconsistency metric, defined by Crosscombe and Lawry [21], between their belief states is greater than a given threshold. We use the same definition of the inconsistency metric, I, presented by Crosscombe and Lawry [21]. In short,

the inconsistency metric captures the fraction of propositions on which two agents hold certain opposing opinions.

In a rewiring event the inconsistency metric between the two nodes belief states $I(B_\alpha, B_\beta)$ is calculated. If this value is less than a given threshold, $I(B_\alpha, B_\beta) < \gamma$, nothing happens while if it is greater than the threshold, $I(B_\alpha, B_\beta) > \gamma$, where $\gamma \in [0,1]$, the link is removed from the set of all links. One of the nodes that was present in the link, $\delta \in \{\alpha, \beta\}$, is chosen uniformly at random. A new link is then created between δ and a node, ϵ, chosen uniformly at random from the set of all nodes who have a consistent belief state, that is $\{\epsilon : I(B_\delta, B_\epsilon) < \gamma\}$.

Next, with regards to opinion adoption events, we assume that agents are more likely to be influenced by agents who share consistent beliefs than by those who do not. In addition, we assume that two nodes with certain conflicting belief states moderate each other in an opinion adoption event by both adopting an uncertain belief state after the event. Using our set notation we propose an equivalent consensus operator, $\circ$, to that proposed by Crosscombe and Lawry [21],

$$B_i \circ B_j = \begin{cases} B_i \cap B_j & \text{if} \quad B_i \cap B_j \neq \emptyset, \\ B_i \cup B_j & \text{if} \quad B_i \cap B_j = \emptyset. \end{cases}$$

Given these assumptions, in an opinion adoption event the inconsistency metric between the two nodes' belief states, $I(B_\alpha, B_\beta)$, is calculated. If this value is greater than the same threshold γ as for the rewiring rates, , $I(B_\alpha, B_\beta) > \gamma$, nothing happens, while if, $I(B_\alpha, B_\beta) < \gamma$, the consensus operator is applied to the pair of belief states B_α and B_β. The result is then assigned to both nodes as their new belief states. We illustrate how the consensus operator works with an example that shows how two nodes with certain belief states prior to the event moderate each other to uncertain belief states after the event. Let the belief state of node i prior to an opinion adoption event be

$$B_i = s^1 \tag{13}$$

and the belief state state of node j prior to the event be

$$B_j = s^2 \tag{14}$$

The consensus operator applied to B_i and B_j is then

$$B_{new} = B_i \circ B_j = \{s^1, s^2\}. \tag{15}$$

This result is then assigned to nodes i and j as their new belief state. The equivalent vector notation for this belief state is

$$B_{new} = \{\frac{1}{2}, \frac{1}{2}\}. \tag{16}$$

This demonstrates how the consensus operator moderates the node's opposing certain beliefs in an opinion adoption event.

We introduce a random state change event in an attempt to model noise in a real-world system. In a random state change event one of the nodes, $\delta \in \{\alpha, \beta\}$, is chosen uniformly at random from the link. An integer, i, is then sampled from the uniform distribution on the set

$\{1,...,n\}$. The nodes opinion about the corresponding proposition, p_i, is then changed to one of the other possible opinions with equal probability. The resultant belief state is then normalised appropriately if required. We present an example of a random state change event using the vector notation. Let,

$$B_{before} = \{0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}\}. \tag{17}$$

In the random state change event the integer 2 is sampled from the uniform distribution on the set $\{1,2,3,4,5\}$. The agent currently holds the opinion of $\frac{1}{2}$ for this proposition. With probability $P = \frac{1}{2}$ the agents opinion about this proposition is changed to 0 and the resultant belief state after the event is given by

$$B_{after} = \{0, 0, 0, \frac{1}{2}, \frac{1}{2}\}. \tag{18}$$

This belief state is valid and therefore does not need normalising. With probability $P = \frac{1}{2}$ the agents opinion is changed to 1 and the resultant belief state after the event is given by

$$B_{after} = \{0, 1, 0, \frac{1}{2}, \frac{1}{2}\}. \tag{19}$$

This belief state is not valid due to the mutual exclusivity of propositions. This means the belief state must be normalised to

$$B_{after} = \{0, 1, 0, 0, 0\} = s^2. \tag{20}$$

Finally, following the work of Mobilla et al,. [13], we include the presence of zealots with the aim of modelling individual failure in a real-world system. A zealot is a node that does not update its belief state in an opinion adoption event or a random state change event.

2.4 Different versions of the model

We have described the formulation of the T.V. model. We arrive at the B.V. model by making a few adjustments to this description. In the B.V. model an agents opinion about a single proposition is represented as a binary, as opposed to a ternary, variable. An agents belief state is then defined as an n-dimensional vector in $\{0,1\}^n$. Agents can only have certain belief states, due to the mutual exclusivity of the propositions, and therefore the quality of an agents belief state is simply given by the quality of the proposition they consider to be the best, q_i. We use the same model for agent interaction and decision rules except that the opinion adoption event is changed slightly. In an opinion adoption event when a link, (α, β), is chosen and $I\left(B_\alpha, B_\beta\right) < \gamma$ a node present in the link, $\delta \in \{\alpha, \beta\}$, is chosen uniformly at random. This node then adopts the belief state of the other node.

3 Analysis

In order to analytically study the effect of uncertainty on the fault tolerance and the noise tolerance of the adaptive voter model we will first reduce the complexity of the T.V. and the B.V. models derived in Chapter 2. We will measure the noise tolerance of the models by; calculating the time taken to reach a consensus state in the presence of noise and determining the critical rewiring rate required to cause full fragmentation in the presence of noise. We will measure

the fault tolerance of the models by determining the critical rewiring rate required to cause full fragmentation in the presence of zealots. We suggest that these metrics are relevant to solving the best-of-n problem in swarm robotics as; the more fragmented the network is the less opportunity there is for the emergence of collective intelligence and the faster the swarm converges on the optimal decision the better.

3.1 Simplified models

In order to simplify the analysis of the models we assume that all propositions have the same deterministic quality and that there is no inconsistency between individual agents belief states meaning that rewiring events and opinion adoption events can occur on any active link. We introduce a further simplification in the opinion adoption event for the T.V. model. When an active link containing an agent with a certain belief state, 0 or 1, and an uncertain belief, $\frac{1}{2}$, is selected only the agent with the uncertain belief state can adopt the opinion of the agent with the certain belief state. In addition, when an active link containing two agents with opposing certain belief states, 0 and 1, is selected, one of the agents is chosen at random to adopt the uncertain belief state. We hope that these simplifications to the opinion adoption event still capture the essential features of the consensus operator proposed by Crosscombe and Lawry [21]. We present a brief description of the simplified model for completeness. We shall refer to the simplified T.V. model as the S.T.V. model and the simplified B.V. model as the S.B.V. model.

3.1.1 S.T.V. model

We consider a network of N nodes, that represent agents, and L undirected links, that represent contacts between these nodes. In the S.T.V. model there are two propositions, A and C, about which the nodes can express an opinion, and three possible belief states 0, $\frac{1}{2}$ and 1. Each node, α , maintains a belief state, $B_\alpha \in \{0, \frac{1}{2}, 1\}$, that encodes their opinion. A belief state of 0 represents the opinion that proposition A is the best. We shall refer to nodes with this belief state as A nodes. A belief state of 1 represents the opinion that proposition C is best. We shall refer to nodes with this belief state as C nodes. A belief state of $\frac{1}{2}$ represents the borderline case where a node is uncertain as to which proposition is the best. We shall refer to nodes with this belief state as B nodes. The network is initialised as a random graph, generated from the configuration model $G(n,p)$, with mean degree $\langle k \rangle = \frac{2L}{N}$ [22]. Each node is assigned an initial belief state by sampling from a uniform distribution on the set $\{0, \frac{1}{2}, 1\}$.

The network is evolved in time through successive update steps. In each update step a link, (α, β), is chosen uniformly at random from the set of all links. If the belief states are equal, $B_\alpha = B_\beta$, then the link is considered to be inert and nothing happens. If the belief states are different, $B_\alpha \neq B_\beta$, then the link is considered to be active and an update event occurs on the chosen link. There are three types of active link which we shall refer to as AB $\left(\{B_\alpha, B_\beta\} = \{0, \frac{1}{2}\}\right)$, AC $\left(\{B_\alpha, B_\beta\} = \{0, 1\}\right)$ and BC $\left(\{B_\alpha, B_\beta\} = \{\frac{1}{2}, 1\}\right)$. In short, AB and BC links represent conflicts between individuals with certain and uncertain beliefs whilst AC links represent conflicts between individuals with opposing certain beliefs. There are two classes of update events, namely rewiring events and opinion adoption events. When an active link is selected, a binary choice is made about which type of event occurs. For AB, AC and BC links, a rewiring event

occurs with probability p_1, p_2 and p_3 respectively whilst an opinion adoption event occurs with probability $(1-p_1)$, $(1-p_2)$ and $(1-p_3)$ respectively. For notational convenience we shall denote the opinion adoption event probabilities as $\phi_1 = (1-p_1)$, $\phi_2 = (1-p_2)$ and $\phi_3 = (1-p_3)$.

In a rewiring event the chosen link, (α, β), is removed from the set of all links. One of the nodes, $\gamma \in \{\alpha, \beta\}$, from the link is chosen uniformly at random. A new link is then created from γ to a node δ where δ is sampled from a uniform distribution on the set of all nodes that share the same belief state as γ, $B_\delta = B_\gamma$. In this model there are three rewiring events: rewiring an AB link, rewiring an AC link and rewiring a BC link. As stated above these events occur with probabilities p_1, p_2 and p_3 respectively.

In an opinion adoption event there are two distinct cases to consider. If the chosen link is an AC link then one of the nodes present in the link, $\gamma \in \{\alpha, \beta\}$, is chosen uniformly at random. The chosen node γ then resolves the conflict in accordance with the consensus operator presented by Crosscombe and Lawry [21] by adopting the uncertain belief state such that $B_\gamma = \frac{1}{2}$. If the link is an AB or a BC link then the node with belief state $\frac{1}{2}$ resolves the conflict by adopting the belief state of the node with the certain belief state 0 or 1. In this model there are three opinion update events: opinion update on an AB link, AC link and BC link. These events occur with probabilities probabilities ϕ_1, ϕ_2 and ϕ_3 respectively.

3.1.2 S.B.V. model

We again consider a network of N nodes, that represent agents, and L bidirectional links, that represent contacts between these nodes. In the S.B.V. model there are three propositions, A, B and C , about which the nodes can express an opinion about and three possible belief states 0, 1 and 2. We opt for a different numbering scheme in order to avoid confusion with the S.T.V. model. Each node, α , maintains a belief state, $B_\alpha \in \{0,1,2\}$, that encodes their opinion. A belief state of 0 represents the opinion that proposition A is the best. We shall refer to nodes with this belief state as A nodes. A belief state of 1 represents the opinion that proposition B is best. We shall refer to nodes with this belief state as B nodes. A belief state of 2 represents the opinion that proposition C is the best. We shall refer to nodes with this belief state as C nodes. The network is initialised as a random graph, generated from the configuration model $G(n,p)$, with mean degree $\langle k \rangle = \frac{2L}{N}$ [22]. Each node is assigned an initial belief state by sampling from a uniform distribution on the set $\{0,1,2\}$.

The network is evolved in the same way as the S.T.V. model however in this model each type of active link, AB, AC and BC, represents a conflict between agents with opposing certain beliefs. The rates for rewiring and opinion adoption events are the same, and the rewiring event on active links is also the same. The only difference in network evolution is in the opinion adoption event. When any of the active links is chosen, one of the nodes present in the link, $\gamma \in \{\alpha, \beta\}$, is chosen uniformly at random. The chosen node γ then resolves the conflict by adopting the belief state of the other node.

3.2 Moment closure methodology

A common problem in modelling the dynamics of a stochastic process on a network is the high dimensionality of the resulting dynamical system [23]. The network models described

above contain N nodes and 3 states. A stochastic process describing the complete time evolution of these states would have dimensionality $\mathcal{O}(3^N)$ at each time step [23]. In general, a dimensionality-reduction technique must be used to proceed with analysis.

A commonly used class of techniques for dimensionality reduction is known as moment closure [23]. As a preliminary we must define what we mean by a network motif. We say that a network A contains a motif B if we can find a subset of nodes $V' \subset V$, where V is the set of nodes of the network A, such that the subgraph of A induced by V' is isomorphic to B [24]. The first step in applying a moment closure technique is to define small motifs of interest in the network, such as adjacent pairs or triplets of nodes. It is then assumed that the dynamics of these motifs are only affected by the state of motifs of the same or lower dimension [23]. It is also assumed that the densities of the identified motifs change smoothly over time [25]. This assumption ignores the stochastic fluctuations in the system that arise due to the probabilistic quality of the events that change the densities of the motifs. Fortunately, this assumption is fairly benign in large networks, $N \gg 1$ [25]. This procedure permits the derivation of a closed system of differential equations that describe the time evolution of these motifs [23].

3.3 Consensus time

This section adapts the method presented by Rogers et al., [26] in order to calculate the time taken to reach consensus in the S.B.V and S.T.V. models. The method is called the 'pair-based proxy', as it based on on a moment-closure assumption that is similar to the pair-approximation [26]. The method involves the derivation of a low-dimensional Markov jump process that approximates the dynamics of the full system by investigating the dynamics of a set of variables representing the number of agents in each state and the number of links connecting agents in given states [20]. We present the derivation for the S.T.V. model defined in 3.1.1, with, for simplicity of illustration, $p_1 = p_2 = p_3 = p$ and $(1-p) = \phi_1$.

We first define a system state vector X that approximates the state of the full system by tracking the number of each node type and each link type,

$$X = \Big[[A] \quad [B] \quad [C] \quad [AA] \quad [BB] \quad [CC] \quad [AB] \quad [AC] \quad [BC] \Big]. \tag{21}$$

where $[X]$ denotes the number of X nodes and $[XY]$ denotes the number of links connecting X nodes to Y nodes. There are 10 possible events that can change the state of the system, X. Specifically, there are 6 rewiring events and 4 opinion adoption events. For each of these events, we determine the probability that the event occurs in a given time step, r_i, and the average net change, Δ_i, this event induces in X. We can then use these rates and net changes to approximate the dynamics of the state vector X with a low-dimensional Markov jump process [20, 26]. In each time step, a jump vector Δ_i is chosen randomly with probability r_i. The system state vector X is then updated as follows

$$X_{i+1} \mapsto X_i + \Delta_i. \tag{22}$$

We proceed by enumerating the possible events, the rates with which they occur and the average net change they induce in X.

3.3.1 Rewiring events

Rewiring events only occur on active links, so we are only concerned with link types AB, AC and BC. For each link type there are two possible rewiring events. Given an AB link: the A node may rewire the link to another A node and the B node may rewire the link to another B node. The rewiring events for AC and BC link types are equivalent to those for the AB link type. The event, the probability of the event occurring and the change in X that the event causes are summarised in Table 1.

Event	Pr(Event)	Change in X
$AB \mapsto AA$	$r_1 = \frac{[AB]}{L}\frac{p}{2}$	$\Delta_1 = [0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0]$
$AB \mapsto BB$	$r_2 = \frac{[AB]}{L}\frac{p}{2}$	$\Delta_3 = [0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0]$
$AC \mapsto AA$	$r_3 = \frac{[AC]}{L}\frac{p}{2}$	$\Delta_3 = [0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0]$
$AC \mapsto CC$	$r_4 = \frac{[AC]}{L}\frac{p}{2}$	$\Delta_4 = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0]$
$BC \mapsto BB$	$r_5 = \frac{[BC]}{L}\frac{p}{2}$	$\Delta_5 = [0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1]$
$BC \mapsto CC$	$r_6 = \frac{[BC]}{L}\frac{p}{2}$	$\Delta_6 = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1]$

Table 1: Summary all of the possible rewiring events, the probabilities with which they occur and the change they induce in X.

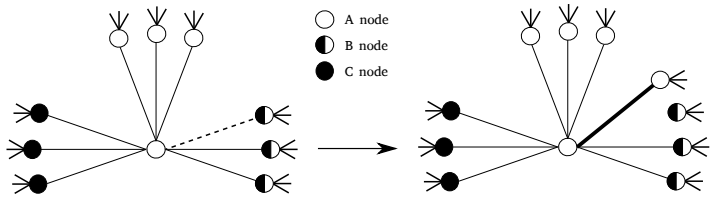


Figure 1: Graphical representation of the event $AB \mapsto AA$. The link on which the update occurs is identified with a dashed line. The new link is identified with a thick black line.

In Table 1, the event $AB \mapsto AA$ is to be interpreted as the event where an A node from an AB link breaks the link and rewires itself to another A node creating a new AA link. Figure 1 depicts the event graphically. The probability of the event $AB \mapsto AA$ occurring is $r_1 = \frac{[AB]}{L}p\frac{1}{2}$. The first term in the expression for r_1 corresponds to the probability of selecting an AB link from the set of L links. The second term in the expression corresponds to the probability of a rewiring event occurring whilst the third term corresponds to an A node being picked to rewire itself to another A node. The change this event induces in X is $\Delta_1 = \begin{bmatrix}0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0\end{bmatrix}$ because the number of AA links, $[AA]$, increases by 1 whilst the number of AB links, $[AB]$ decreases by 1. The other events in Table 1 are to be interpreted analogously.

3.3.2 Opinion adoption events

Opinion adoption events also only occur on active links so we are only concerned with link types AB, AC and BC. There are four possible opinion adoption events that can occur on these links. On an AB or a BC link it is only the B node that can adopt the belief state of the A or C node. In contrast, on an AC link, one of the nodes is chosen uniformly at random to adopt the belief

state of a B node. We shall provide a detailed description of one of the opinion adoption events as the rest can be described in the same way. We shall focus on the opinion adoption event where an A node present in an AC link adopts the belief state of a B node. The probability of this event occurring is given by,

$$\Pr(\text{A adopts B}) \quad = \quad r_7 \quad = \quad (1-p)\frac{[AC]}{L}\frac{1}{2}. \tag{23}$$

The first term in (23) corresponds to the probability of an opinion adoption event occurring. The second term corresponds to picking an AC link from the set of all links. The third term corresponds to picking an A node to adopt the belief state of a B node.

In order to determine the change in X induced by this event we must consider what happens to all the links connected to the A node when it changes belief state to become a B node. It will be helpful to refer to Figure 2 for the following derivation.

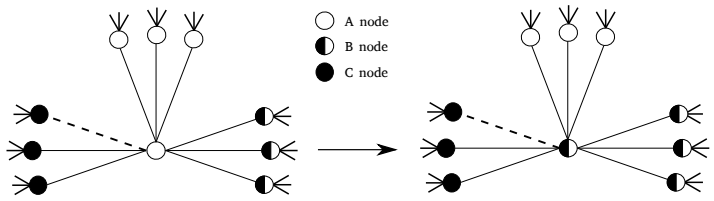


Figure 2: Graphical representation of the event in which an A node adopts the belief state of a B node. The link on which the update occurs is identified with a dashed line.

On average a typical A node has

$$k_A = \frac{2[AA]+[AB]+[AC]}{[A]} \tag{24}$$

neighbours. To understand why, consider each node type in turn. We shall begin by counting the number of B neighbours a typical A node has. Each AB link in the network contains an A node and a B node. The number of AB links, $[AB]$, therefore tells us the number of A nodes that are connected to B nodes. The expression $[AB]/[A]$ is then the average number of B nodes that each A node is connected to. So we expect the typical A node to have $[AB]/[A]$ neighbours of type B. The same reasoning follows for the expected number of C neighbours for a typical A node, which is therefore given by the expression $[AC]/[A]$. We must be a bit more careful when considering the number of A neighbours a typical A node has. Each AA link contains 2 A nodes. This means that the expression $2[AA]$ tells us the number of A nodes that are connected to A nodes in the network. We therefore expect a typical A node to have $2[AA]/[A]$ neighbours of type A. We know that the node under consideration must have at least one C neighbour. Using expression (24) we deduce that the average number of additional A, B and C neighbours of an

opinion adoption A node in an AC link are

$$\text{Additional A neighbours} = \Big(\frac{k_A-1}{k_A}\Big)\frac{2[AA]}{[A]} = \Big(1-\frac{1}{k_A}\Big)\frac{2[AA]}{[A]} = A_A, \tag{25a}$$

$$\text{Additional B neighbours} = \Big(\frac{k_A-1}{k_A}\Big)\frac{[AB]}{[A]} = \Big(1-\frac{1}{k_A}\Big)\frac{[AB]}{[A]} = A_B, \tag{25b}$$

$$\text{Additional C neighbours} = \Big(\frac{k_A-1}{k_A}\Big)\frac{[AC]}{[A]} = \Big(1-\frac{1}{k_A}\Big)\frac{[AC]}{[A]} = A_C \tag{25c}$$

It is clear that this event causes the number of A nodes in the system to decrease by 1, the number of B nodes to increase by 1 and the number of C nodes to remain constant. We will use the expressions derived above to show what changes the event induces in $[AA]$,$[BB]$,$[CC]$,$[AB]$,$[AC]$ and $[BC]$. We quantify the number of each link type connected to the A node before and after the event. We then determine the change in the number of each link type induced by the event. These calculations are summarised in the Table 2.

Link type	Number before	Number after	Difference
$[AA]$	A_A	0	$-A_A$
$[BB]$	0	A_B	A_B
$[CC]$	0	0	0
$[AB]$	A_B	A_A	$A_A - A_B$
$[AC]$	$1+A_C$	0	$-(1+A_C)$
$[BC]$	0	$1+A_C$	$1+A_C$

Table 2: Changes in the number of link types $[AA]$,$[BB]$,$[CC]$,$[AB]$,$[AC]$ and $[BC]$ induced by the event where an A node adopts the belief state of a B node.

We now have the necessary parts to describe the typical change induced in the state vector X when an A node from an AC link adopts the belief state of a B node

$$\Delta_7 = \begin{bmatrix} -1 & 1 & 0 & -A_A & A_B & 0 & A_A - A_B & -(1+A_C) & 1+A_C \end{bmatrix}. \tag{26}$$

The three remaining opinion adoption events are: a C node from an AC link adopts the belief state of a B node, a B node on an AB link adopts the belief state of an A node and a B node on a BC link adopts the belief state of a C node. We use the same method as before to calculate the probability of these events occurring and the change they induce in X when they occur. On average a typical B node has

$$k_B = \frac{2[BB]+[AB]+[BC]}{[B]}, \tag{27}$$

neighbours, whilst on average a typical C node has

$$k_C = \frac{2[CC]+[AC]+[BC]}{[C]}. \tag{28}$$

neighbours. We omit the rest of the calculations and just present the results in Table 3.

AC link	AB link	BC link
$C \mapsto B$	$B \mapsto A$	$B \mapsto C$
$r_8 = (1-p)\frac{[AC]}{2L}$	$r_9 = (1-p)\frac{[AB]}{2L}$	$r_{10} = (1-p)\frac{[BC]}{2L}$
$\Delta_8 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \\ C_B \\ -C_C \\ 1+C_A \\ -(1+C_A) \\ C_C - C_B \end{bmatrix}$	$\Delta_9 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1+B_A \\ -B_B \\ 0 \\ B_B-(1+B_A) \\ B_C \\ -B_C \end{bmatrix}$	$\Delta_{10} = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \\ -B_B \\ 1+B_C \\ -B_A \\ B_A \\ B_B-(1+B_C) \end{bmatrix}$

Table 3: Summary of the remaining opinion adoption events, the probabilities with which they occur and the change they induce in X where B_A, B_B, B_C, C_A, C_B and C_C are to be interpreted in the same way as A_A, A_B and A_C.

Rogers and Gross [20] caution that there is an implicit assumption in 'pair-based proxy'. When we approximated the number of additional A, B and C neighbours an updated A node had we used the expressions (25a), (25b) and (25c). These expressions provide an estimate for a typical A node. In some cases, the fact that the A node under consideration has at least one A, B or C neighbour makes the node atypical [20]. An example of such a case could be when there is only 1 AC link remaining in the network. Rogers and Gross [20] state that this assumption is equivalent to the pair-approximation. They claim that this assumption is justified, even though there is no bound on the size of the error caused, as it generally works well in the voter model when $(1-p)$ is not too large and k is not too small [20, 27]. Following the work of Rogers and Gross [20] we encapsulate the update vectors Δ_i in a stoichiometric matrix

$$\Delta = \begin{bmatrix} \boldsymbol{\Delta}_1 & : & \boldsymbol{\Delta}_2 & : & \boldsymbol{\Delta}_3 & : & \boldsymbol{\Delta}_4 & : & \boldsymbol{\Delta}_5 & : & \boldsymbol{\Delta}_6 & : & \boldsymbol{\Delta}_7 & : & \boldsymbol{\Delta}_8 & : & \boldsymbol{\Delta}_9 & : & \boldsymbol{\Delta}_{10} \end{bmatrix} \tag{29}$$

and the event probabilities in a column vector

$$\mathbf{r} = \frac{1}{2L} \begin{bmatrix} [AB]p \\ [AB]p \\ [AC]p \\ [AC]p \\ [BC]p \\ [BC]p \\ (1-p)[AC] \\ (1-p)[AC] \\ (1-p)[AB] \\ (1-p)[BC] \end{bmatrix}. \tag{30}$$

We can now provide an analytical description of the jump process in the limit of large N. We

can rescale the variables such that,

$$a = \frac{[A]}{N} \quad b = \frac{[B]}{N} \quad c = \frac{[C]}{N} \tag{31}$$

$$aa = \frac{[AA]}{L} \quad bb = \frac{[BB]}{L} \quad cc = \frac{[CC]}{L} \tag{32}$$

$$ab = \frac{[AB]}{L} \quad ac = \frac{[AC]}{L} \quad bc = \frac{[BC]}{L} \tag{33}$$

for $N \gg 1$ and $L \gg 1$. The rescaled state vector $\boldsymbol{x}$,

$$\boldsymbol{x} = \begin{bmatrix} a & b & c & aa & bb & cc & ab & ac & bc \end{bmatrix} \tag{34}$$

is a realisation of a density dependent Markov process that satisfies the condition of Kurtz's theorem [26, 28]. We can apply this theorem to obtain a Langevin equation for $\boldsymbol{x}$ that describes the mesoscopic behaviour of the network in the limit of large N

$$\frac{d}{dt}\boldsymbol{x} = \Delta \boldsymbol{r} + \frac{1}{\sqrt{N}}\boldsymbol{\eta}(t), \tag{35}$$

where $\boldsymbol{\eta}(t)$ is a multiplicative noise term, such that

$$\left\langle \eta_i\left(t\right), \eta_j\left(t'\right) \right\rangle = \delta\left(t - t'\right) B_{ij}, \tag{36}$$

with the noise correlation matrix B given by,

$$B_{ij} = \sum_k B_{ik} r_k B_{kj}. \tag{37}$$

We employ the 'pair-based proxy' to obtain another Langevin equation that describes the mesoscopic behaviour of the S.B.V. model in the limit of large N.

3.3.3 Random state change event

We claim that it is pointless to investigate the time taken to reach a consensus in the presence of zealots because consensus can only occur if there is one zealot and all the other nodes in the network adopt the belief state of this zealot. It is unlikely that this consensus would be optimal and that there would only be one zealot in the population in a real-world system. We therefore choose not to investigate the presence of zealots.

Instead we focus on investigating how the presence of noise in the network affects the consensus time in the S.T.V. and S.B.V models. As in Section 3.3 we only detail the derivation for the S.T.V. model as the derivation for the S.B.V. model follows the same procedure. There are 6 possible random state change events in the S.T.V model: an A node changing into a B or a C node, a B node changing into an A or a C node and a C node changing into an A or a B node. These events can occur on any node. Following the same procedure as used in the previous section we find the probability of the event occurring and the change induced in X for each of these random state change events. The results are summarised in Table 4 with R representing the probability of a random state change event occurring,

Event	Pr(Event)	Change in X
$A \mapsto B$	$r_{11} = R\frac{[A]}{2N}$	$\Delta_{11} = \begin{bmatrix} -1 & 1 & 0 & -\frac{2[AA]}{[A]} & \frac{[AB]}{[A]} & 0 & \frac{2[AA]-[AB]}{[A]} & -\frac{[AC]}{[A]} & \frac{[AC]}{[A]} \end{bmatrix}$
$A \mapsto C$	$r_{12} = R\frac{[A]}{2N}$	$\Delta_{12} = \begin{bmatrix} -1 & 0 & 1 & -\frac{2[AA]}{[A]} & 0 & \frac{[AC]}{[A]} & -\frac{[AB]}{[A]} & \frac{2[AA]-[AC]}{[A]} & \frac{[AB]}{[A]} \end{bmatrix}$
$B \mapsto A$	$r_{13} = R\frac{[B]}{2N}$	$\Delta_{13} = \begin{bmatrix} 1 & -1 & 0 & \frac{[AB]}{[B]} & -\frac{2[BB]}{[B]} & 0 & \frac{2[BB]-[AB]}{[B]} & \frac{[BC]}{[B]} & -\frac{[BC]}{[B]} \end{bmatrix}$
$B \mapsto C$	$r_{14} = R\frac{[B]}{2N}$	$\Delta_{14} = \begin{bmatrix} 0 & -1 & 1 & 0 & -\frac{2[BB]}{[B]} & \frac{[BC]}{[B]} & -\frac{[AB]}{[A]} & \frac{[AB]}{[A]} & \frac{2[BB]-[BC]}{[B]} \end{bmatrix}$
$C \mapsto A$	$r_{15} = R\frac{[C]}{2N}$	$\Delta_{15} = \begin{bmatrix} 1 & 0 & -1 & \frac{[AC]}{[C]} & 0 & -\frac{2[CC]}{[C]} & \frac{[BC]}{[C]} & \frac{2[CC]-[AC]}{[C]} & -\frac{[BC]}{[C]} \end{bmatrix}$
$C \mapsto B$	$r_{16} = R\frac{[C]}{2N}$	$\Delta_{16} = \begin{bmatrix} 0 & 1 & -1 & 0 & \frac{[BC]}{[C]} & -\frac{2[CC]}{[C]} & \frac{[AC]}{[C]} & -\frac{[AC]}{[C]} & \frac{2[CC]-[BC]}{[C]} \end{bmatrix}$

Table 4: Summary all of the possible random state change events, the probabilities with which they occur and the change they induce in X.

Using the same rescaling as in Section 3.3.2 we arrive at another Langevin equation that describes the mesoscopic behaviour of the S.T.V. model in the limit of large N in the presence of noise. We repeat this process for the S.B.V. model. We refer to these extended models as the noisy S.T.V. model and the noisy S.B.V. model.

3.3.4 Results

In the limit of large population size, also known as the thermodynamic limit where $N \gg 1$ and $L \gg 1$ with $\frac{L}{N}$ constant, the stochastic fluctuations become negligible and the system tends towards deterministic behaviour [26]. This means that equation (35) can be reduced to a deterministic set of differential equations,

$$\frac{d}{dt}\boldsymbol{x} = \Delta\boldsymbol{r}. \tag{38}$$

Despite the appeal of dealing with a set of ODE's as opposed to a set of SDE's it has been shown by Toral and Tessone [29] that finite-size effects can produce dynamics that are not predicted in the thermodynamic limit. In an attempt to avoid neglecting finite-size effects we choose to proceed with the original set of SDE's (35). A common technique for gaining analytical traction with such an equation is to derive a first-order approximation for the effects of noise by linearising the Langevin equation using a procedure equivalent to the van Kampen system size expansion [26]. It is known that this approximation fails to reproduce some features of systems with an absorbing state [30]. In light of this, Di Patti et al., [30] present an alternative approximation that converges to the exact solution, attained from a numerical implementation of the Gillespie algorithm, for a system with an absorbing state.

Due to time constraints, we could not investigate the application of this proposed expansion to our models. Instead, we obtain results using a numerical implementation of the Gillespie algorithm that generates statistically correct trajectories [31].

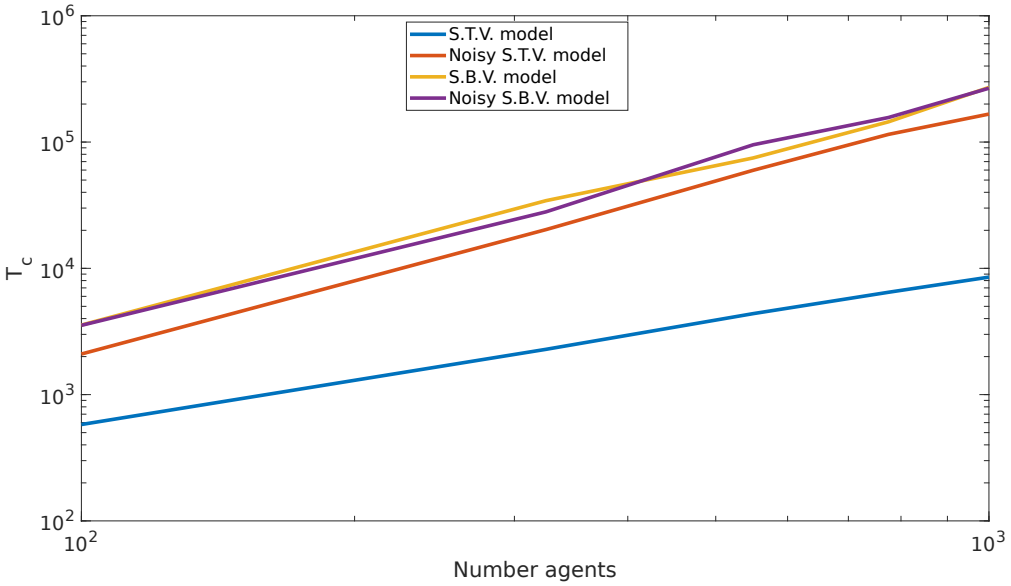


Figure 3: Scaling behaviour of the mean time to consensus, T_c, as a function of network size N for the S.T.V and B.T.V. models with and without the presence of noise. The results were generated by averaging over 100 realisations with: $\langle k \rangle = 10$, $p = 0.1$ and $R = 0.01$

Figure 3 depicts preliminary results on the scaling behaviour of the mean time to consensus, T_c, as a function of network size N, for the derived models. It can be seen that all of the models exhibit linear scaling in N. Rogers and Gross [20] analytically derive a linear scaling,

$$T_c = \frac{N \log(2)}{(\phi_* - \phi)} \tag{39}$$

where

$$\phi_* = \frac{(k-2)}{(k-1)}, \tag{40}$$

for the traditional adaptive voter model, which is equivalent to our S.B.V. model. This suggests that our results, obtained from numerical simulation, are accurate. Figure 3 shows that the S.T.V. model takes less time to converge on a consensus state than the S.B.V. model for networks with a population size of $N \in \{10^2, \ldots, 10^3\}$. In addition, it shows that the noisy S.T.V. model reaches consensus faster than the noisy S.B.V. model for networks with a population size of $N \in \{10^2, \ldots, 10^3\}$.

Interestingly, it seems that the S.T.V. model is more affected by the presence of noise than the S.B.V. model. This could potentially be explained with reference to the symmetry in the models. The addition of noise imposes a certain symmetry on the network as it affects each node with the same probability. As the S.B.V. model already possess symmetry in the dynamics of the node and link densities, due to the equivalence of the belief states, the addition of extra symmetry from the presence of noise has little effect on the network dynamics. On the other hand, the S.T.V. model contains some asymmetry due to the one way opinion adoption dynamics on AB and BC links. The symmetry imposed by the presence of noise therefore has a significant effect on the network dynamics. It could be argued that the scaling behaviour of the S.T.V. model will approach that of the S.B.V. model for some critical value of the rewiring rate $R = R*$. These

results suggest that the inclusion of uncertainty improves the noise tolerance of the adaptive voter model when the level of noise is below some critical value $R = R*$.

3.4 Fragmentation threshold

This section extends the methodology presented by Bohme and Gross [19] in order to determine the fragmentation transition in a three-state adaptive voter model that includes an uncertain belief state. The subsequent derivation uses the S.T.V. model. In particular we will refer to Section 3 of the work by Bohme and Gross [19], which focuses on fragmentation transitions in a three-state adaptive voter model without an uncertain belief state which is equivalent to the S.B.V. model.

The traditional adaptive voter model contains just two belief states. It has been shown that the model exhibits a fragmentation transition which separates an active regime from a fragmented regime [19]. The active regime is characterised by the persistence of active links in the long-term dynamics [19]. Stochastic fluctuations ultimately drive the system to an absorbing consensus state in which all nodes share the same belief state [19]. The fragmented regime is characterised by the emergence of disconnected components in which all nodes share the same belief state. These disconnected components are internally in consensus [19]. The question we hope to address is whether the addition of an uncertain truth state can prevent or delay the onset of fragmentation.

Due to the presence of a number of different types of active links in the three-state adaptive voter models, whether with or without an uncertain belief state, can evolve to different types of fragmented states. These can be classified as partially fragmented states and a fully fragmented states. In a partially fragmented state the network is comprised of two disconnected components that we call active and fragmented. The fragmented component only contains nodes with the same belief state, such as all A nodes or all C nodes. This component can therefore be considered as static as it is in internal consensus. The active component contains a mixture of nodes with the remaining two belief states. The dynamics in the active component will persist until internal consensus is reached in each component of the network. In a network with three belief states there are three different partially fragmented states. We shall refer to the state in which the fragmented component contains A nodes and the active component contains a mixture of B and C nodes as the A partially fragmented state. The B and C partially fragmented states are defined analogously. In a fully fragmented state the network is comprised of three disconnected components where each component only contains nodes with the same belief state [19].

Bohme and Gross [19] show how the fragmentation transition in the three-state voter model without an uncertain belief state depends on the rewiring rates (p_1, p_2, p_3) and the active link densities $\left(\rho_{AB}, \rho_{AC}, \rho_{BC}\right)$. We want to explore how exchanging one of the certain belief states with an uncertain belief state affects the fragmentation transition. In particular we want to know whether the inclusion of uncertainty makes the network more robust by increasing the rewiring rates required to cause fragmentation. This is important because consensus cannot be reached once the network fragments.

In order to calculate how the fragmentation transition for each of the possible fragmented states depends on the rewiring rates and the link densities we follow the method employed by

Bohme and Gross [19]. As with the previous section the method exploits the moment closure technique by first defining motifs of interest, and then deriving evolution equations for these motifs close to the fragmentation transition. In the context of our S.T.V. model we consider the simpler active motif base presented by Bohme and Gross [18]. This active motif base rests on the assumption that all nodes in the network have the mean degree $\langle k \rangle$. This assumption seems questionable as rewiring events will destroy this degree regularity [18] however Bohme and Gross [18] show that it does yield better results than those obtained through pair-approximation. The spider motif base presented by Bohme and Gross [18] more accurately models the actual degree distribution of the network but it increases the complexity of the derived differential equations. We therefore choose to sacrifice accuracy in order to simplify the analysis.

We begin by considering partial fragmentation, in the S.T.V. model, with respect to A nodes. This refers to the case where the fragmented component only contains nodes with belief state 0. We investigate the system close to the fragmentation transition, where we assume that there are two almost disconnected components. One of these components contains only A nodes, which we shall refer to as the A component, whilst the other contains a mixture of B and C nodes, which we shall refer to as the BC component. We then determine whether or not the A partially fragmented state is stable, such that an A partially fragmented state is reached, or unstable, such that the system avoids entering this state [18]. We assume that close to the fragmentation transition these two components are connected by q-fans. Following the method set out by Bohme and Gross [19], we define q-fans as a collection of q active links, AB and AC, connected to a single A node. We consider specific kinds of BC links to be active but do not explicitly include them in the motifs for the sake of simplicity. This will be explained in more detail later in the derivation.

In order to simplify the derivation we consider the case when $\langle k \rangle = 3$, and we assume that every node, α, has degree, d, equal to the mean degree, $\langle k \rangle$, that is $d_\alpha = \langle k \rangle$. Given that $\langle k \rangle = 3$ and that each A node must be connected by at least 1 link to the A component and each B and C node must be connected by at least 1 link to the BC component we can consider only 1 and 2-fans. We begin by enumerating seven relevant 1 and 2-fans in Figure 4, where we use the same node labelling as Figure 2 from Section 3.2 . In this Figure, the A component is represented by the ellipse with either one or two A nodes inside it whilst the BC component is represented by the ellipse with a B and a C node inside it. The transition rules for these 1 and 2-fans are depicted in Figures 6, 7 and 8. In these figures we choose to represent q-fan types without explicitly showing the inert links that are present. We provide the complete picture and a full description for the transition rules governing the type-1 q-fan, in order to clarify how the other transition rules are to be interpreted. In the rest of this section we shall define: $(1-p_1) = \phi_1$, $(1-p_2) = \phi_2$ and $(1-p_3) = \phi_3$. The type-1 q-fan is a 1-fan as it contains 1 active AB link. Each node is assumed to have degree $d_\alpha = 3$ so the A node is connected to two more nodes that are in the A component and the B node is connected to two more nodes that are in the BC component.

There are two possible events that can occur on this active AB link. The fact that there is only one active AB link is represented by the $(\times 1)$ in Figure 5. The first event illustrated in Figure 5 is an opinion adoption event that involves the B node adopting the belief state of the A

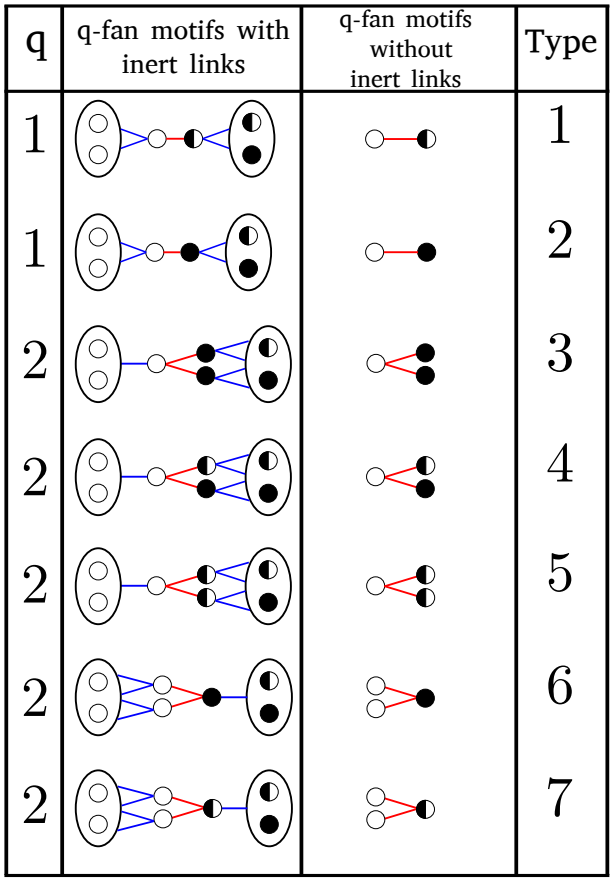


Figure 4: Enumeration of 1 and 2-fans close to the A partial fragmentation threshold. Active links are coloured red and inert links are coloured blue.

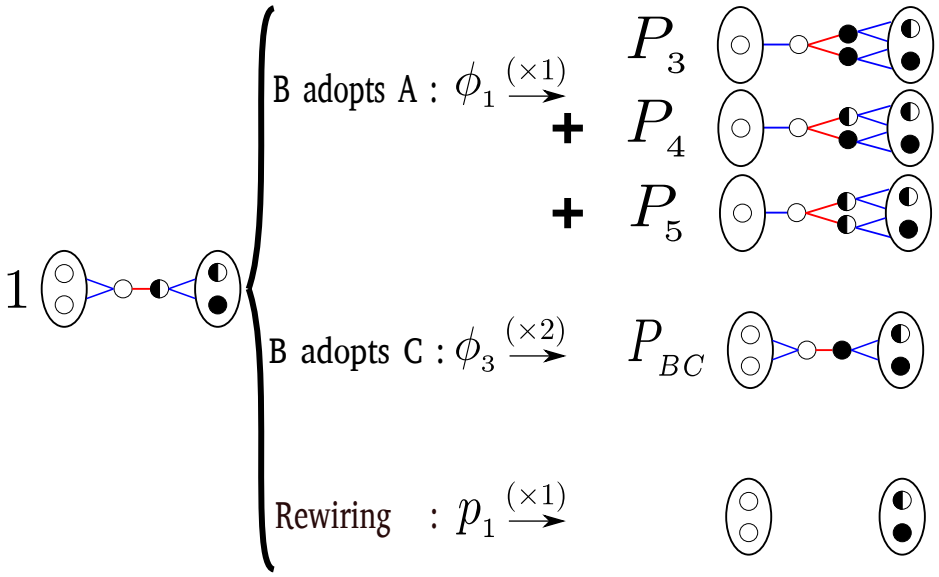


Figure 5: Complete picture of the transition rules governing the type 1 q-fan. Active links are coloured red and inert links are coloured blue.

node. This event occurs with probability ϕ_1. When this event occurs the AB active link becomes an inert AA link. The B node that has now become an A node is now connected to one A node from the A component and it is still connected to two nodes from the BC component. In order to determine which 2-fan this event produces we need to know the type of these two nodes.

We know that they can only be B or C nodes but in order to decide which we must calculate the probability that a B node is connected to another B node and the probability that a B node is connected to a C node. We assume that the active link densities ρ_{AB} and ρ_{AC} vanish close to the fragmentation threshold as there a very small number of them in comparison with AA, BB, CC and BC links. On the other hand, the active link density ρ_{BC} and the inactive link density ρ_{BB} are assumed to be finite. This means that

$$\Pr(\text{B connected to B}) = \frac{\rho_{BB}}{\rho_{BC} + \rho_{BB}} = P_{BB}, \tag{41a}$$

$$\Pr(\text{B connected to C}) = \frac{\rho_{BC}}{\rho_{BC} + \rho_{BB}} = P_{BC}. \tag{41b}$$

Using (41a) and (41b) we can calculate which 2-fans are created by this opinion adoption event and the probability with which they are created. A type-3 q-fan is created if the B node was connected to two C nodes. A type-4 q-fan is created if the B node was connected to one B node and one C node, and a type-5 q-fan is created if the B node was connected to two B nodes. The probabilities with which these 2-fans are created, given that the opinion adoption event happens, are given by:

$$\Pr(\text{Type 3 created}) = \left(\frac{\rho_{BC}}{\rho_{BC} + \rho_{BB}}\right)^2 = P_3, \tag{42a}$$

$$\Pr(\text{Type 4 created}) = \frac{2\rho_{BC}\rho_{BB}}{\left(\rho_{BC} + \rho_{BB}\right)^2} = P_4, \tag{42b}$$

$$\Pr(\text{Type 5 created}) = \left(\frac{\rho_{BB}}{\rho_{BC} + \rho_{BB}}\right)^2 = P_5. \tag{42c}$$

The second event that can occur on the active AB link is a rewiring event, which occurs with probability p_1. When a rewiring event occurs one type-1 q-fan is lost and no new q-fans are created. In the model presented by Bohme and Gross [19], B nodes can adopt the opinion of C nodes and C nodes can adopt the opinion of B nodes with equal probability p_3. Due to this symmetry it is reasonable to assume that ratio of B to C nodes in the BC component will remain relatively constant over time. This justifies the assumption that BC links can be considered inert. In our model only B nodes can adopt the opinion of C nodes when an opinion adoption event occurs on a BC link. This asymmetry suggests that the ratio of B to C nodes in the BC component is likely to decrease over time. In an attempt to account for this we consider BC links connecting B nodes to the BC component to be active.

Given this consideration, the third event that can occur for this type-1 q-fan is an opinion adoption event on a BC link connecting the B node to the BC component. In this event the B node adopts the belief state of a C node. The ($\times 2$) represents the fact that there are two links connecting the B node to the BC component. The P_{BC} term represents the probability that one of those links is a BC link and the ϕ_3 term represents the probability that the B node adopts the

belief state of the C node when an opinion adoption event occurs on a BC link. The transition rules and rates for the rest of the q-fans are depicted in Figures 6, 7 and 8.

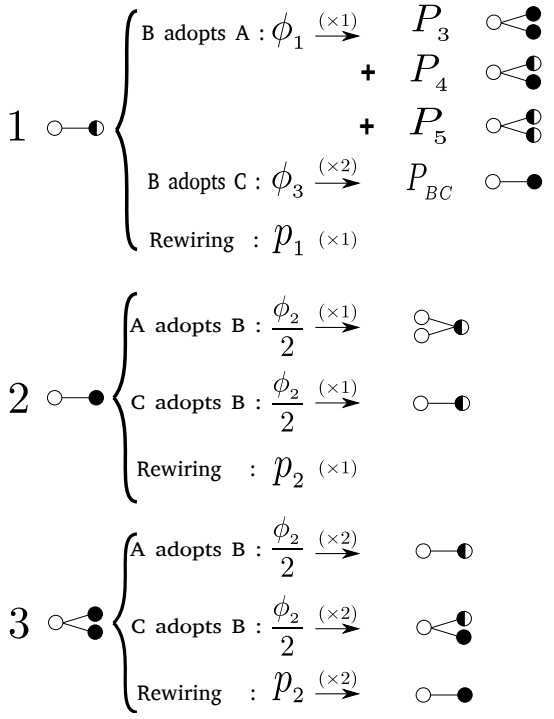


Figure 6: Transition rules for type 1,2 and 3 q-fans close to the A partial fragmentation threshold.

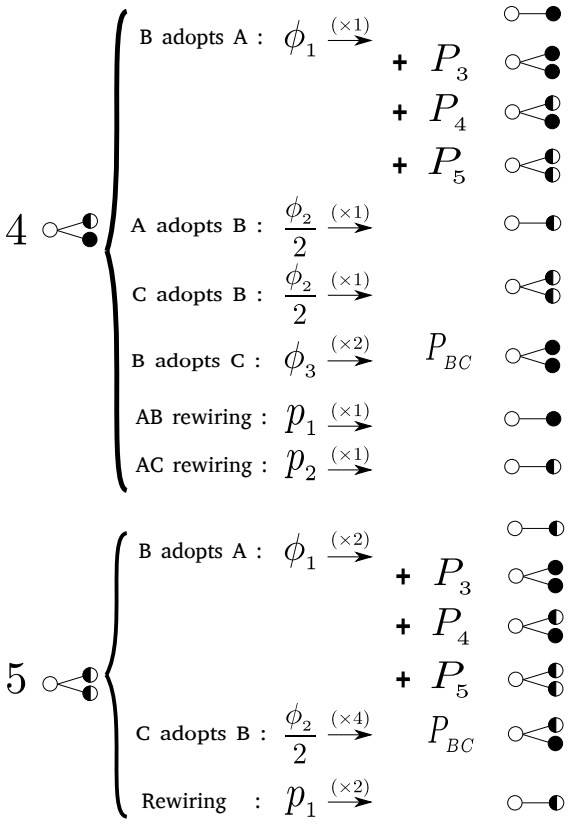


Figure 7: Transition rules for type 4 and 5 q-fans close to the A partial fragmentation threshold.

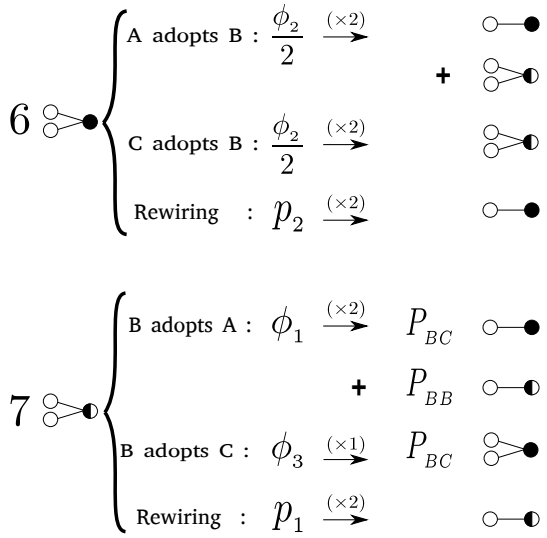


Figure 8: Transition rules for type 6 and 7 q-fans close to the A partial fragmentation threshold.

Using these transition rules we can derive a system of differential equations governing the dynamics of these q-fans. We define,

$$\boldsymbol{X} = \Big([1] \quad [2] \quad [3] \quad [4] \quad [5] \quad [6] \quad [7] \Big), \tag{43}$$

where $[z]$ represents the density of q-fan type z. With this we can now write,

$$\frac{d}{dt}\boldsymbol{X} = \begin{bmatrix} -\left(1+\phi_3 P_{BC}\right)[1] + \frac{\phi_2}{2}[2] + \phi_2[3] + \left(\frac{\phi_2}{2}+p_2\right)[4] + \left(\phi_1+2p_1\right)[5] + \left(\phi_1 P_{BB}+2p_1\right)[7] \\ \phi_3[1] - [2] + 2p_2[3] + \left(\phi_1+p_1\right)[4] + \left(\phi_2+2p_2\right)[6] + \left(\phi_1 P_{BC}\right)[7] \\ \left(\phi_1 P_3\right)[1] - 2[3] + \left(\phi_1 P_3 + \phi_3 P_{BC}\right)[4] + \phi_1 P_3[5] \\ \left(\phi_1 P_4\right)[1] + \left(\phi_2\right)[2] - \left(1+p_1+\phi_3+\phi_1\left(1-P_4\right)\right)[4] - \left(\phi_1 P_4 + 2\phi_3 P_{BC}\right)[5] \\ \left(\phi_1 P_5\right)[1] + \left(\frac{\phi_2}{2}+\phi_1 P_5\right)[4] - \left(2p_1 + 2\phi_3 P_{BC} + \phi_1\left(1-P_5\right)\right)[5] \\ -2[6] + \phi_3 P_{BC}[7] \\ \frac{\phi_2}{2}[2] + 2\phi_2[6] - \left(2+\phi_3 P_{BC}\right)[7] \end{bmatrix} \tag{44}$$

This dynamical system describes how the density of 1 and 2-fans evolves over time. The densities of these active motifs will vanish when the system reaches the A partially fragmented state. In an attempt to provide some bounds on when A partial fragmentation occurs, Bohme and Gross [19] investigate two regimes: $\rho_{BC} = 0$ and $\rho_{BC} = \rho_{BCmax}$. The value for ρ_{BCmax}

$$\rho_{BCmax} = \frac{k-1}{2k}, \tag{45}$$

is obtained through a moment closure approach [19]. In this approach Bohme and Gross [19] derive an evolution equation for the density of BC links in the active BC component using the pair-approximation. They proceed by using this equation to derive an expression for the maximal density of BC links in the BC component in the steady state. In the S.T.V. model the evolution equation for the density of BC links in the active BC component, derived using the pair-approximation, has no additive terms and therefore the density of BC links in the active BC component will decay to 0 over time. In light of this, we only investigate the regime $\rho_{BC} = 0$.

In this regime, $\rho_{BC} = 0$, all of the BC links vanish and the network is reduced to two uncoupled two-state components A-B and A-C [19]. A partial fragmentation will only occur if fragmentation occurs in both of these separate components. The stability of the A partially fragmented state can be analysed by linearising the dynamical system about the trivial equilibrium point where the densities of all the 1 and 2-fans are equal to zero,

$$\boldsymbol{X}^* = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{46}$$

The Jacobian matrix of the linearised dynamical system is given by

$$\boldsymbol{J} = \begin{bmatrix} -\left(1+\phi_3 P_{BC}\right) & \frac{\phi_2}{2} & \phi_2 & \left(\frac{\phi_2}{2}+p_2\right) & (\phi_1+2p_1) & 0 & \left(\phi_1 P_{BB}+2p_1\right) \\ \phi_3 & -1 & 2p_2 & (\phi_1+p_1) & 0 & (\phi_2+2p_2) & \left(\phi_1 P_{BC}\right) \\ \left(\phi_1 P_3\right) & 0 & -2 & \left(\phi_1 P_3+\phi_3 P_{BC}\right) & \phi_1 P_3 & 0 & 0 \\ \left(\phi_1 P_4\right) & 0 & (\phi_2) & -\left(1+p_1+\phi_3+\phi_1\left(1-P_4\right)\right) & -\left(\phi_1 P_4+2\phi_3 P_{BC}\right) & 0 & 0 \\ \left(\phi_1 P_5\right) & 0 & 0 & \left(\frac{\phi_2}{2}+\phi_1 P_5\right) & -\left(2p_1+2\phi_3 P_{BC}+\phi_1\left(1-P_5\right)\right) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 & \phi_3 P_{BC} \\ 0 & \frac{\phi_2}{2} & 0 & 0 & 0 & 2\phi_2 & -\left(2+\phi_3 P_{BC}\right) \end{bmatrix}.$$

The A partially fragmented state is stable when all the eigenvalues of $\boldsymbol{J}$ have negative real part [32]. A bifurcation occurs when at least one of the eigenvalues attains a positive real part. We assume that the occurrence of a bifurcation in our simplified, low dimensional model of the system indicates the occurrence of a fragmentation transition in the full system. We therefore define the condition for the fragmentation transition (the condition for a bifurcation) as $\lambda(\boldsymbol{J}) = 0$ where $\lambda(\boldsymbol{J})$ is the leading eigenvalue of $\boldsymbol{J}$ [19]. This means the fragmentation transition depends on the parameters p_1, p_2, ρ_{BB} and ρ_{BC}. As stated before we have assumed that $\rho_{AB} = 0$ and $\rho_{AC} = 0$. In addition we are considering the regime where $\rho_{BC} = 0$. We choose to set the remaining link densities ρ_{BB}, ρ_{CC} and ρ_{AA} to $\frac{1}{2}$. We numerically calculate $\lambda(\boldsymbol{J})$ in the p_1, p_2 parameter space to produce a two parameter bifurcation diagram, shown in Figure 9

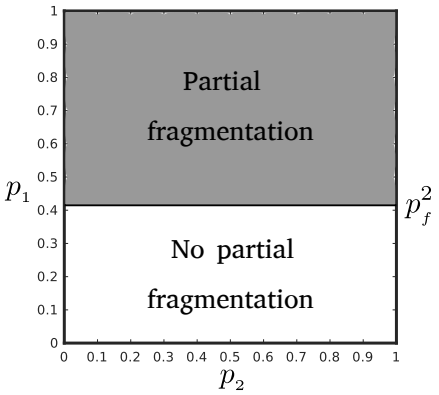


Figure 9: A partial fragmentation phase diagram for the S.T.V. model.

We shall refer to all such bifurcation diagrams as phase diagrams in what follows as we are currently only interested in whether or not a bifurcation occurs and not what kind of bifurcation occurs. We can identify two distinct regions in the phase diagram, Figure 9. In the partial fragmentation region, the grey shaded area, the network reaches an A partially fragmented state for any value of p_3. In the no partial fragmentation region, the white shaded area, the final state of the system depends on the value of p_3. The system transitions to an A partially fragmented state when $p_1 > p_f^2$.

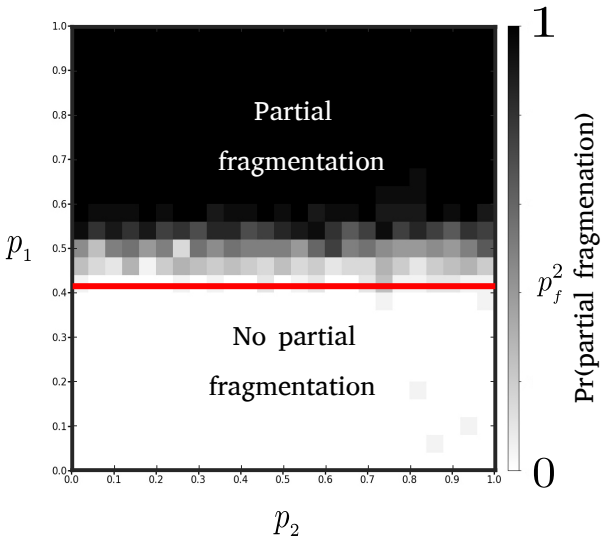


Figure 10: Numerical phase diagram for A partial fragmentation in the S.T.V. model. Colour-coded is the probability that the network reaches a partially fragmented state with black representing a probability of 1 and white representing a probability of 0. Numerical results were obtained by averaging over 10 realisations with $N = 1000$, $\langle k \rangle = 3$ and $p_3 = 0$. Analytical result depicted with thick red line.

Figure 10 demonstrates good agreement between the analytical results and numerical results obtained by agent-based simulation of the S.T.V. model, with a large network size of $N = 1000$. The symmetry in the three-state model without uncertainty, the S.B.V. model, presented by Bohme and Gross [19] permits the derivation of a block-structured Jacobian matrix that governs the stability of the A partially fragmented state. Furthermore, the structure of this

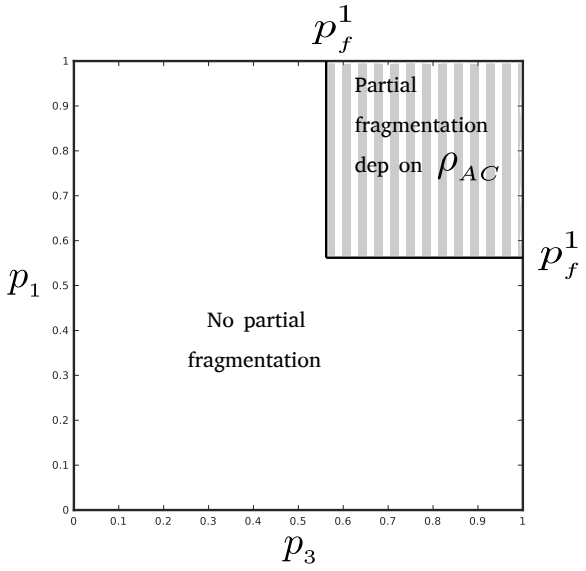


Figure 11: B partial fragmentation phase diagram for the S.T.V. model. Partial fragmentation with respect to B never occurs in the white shaded area and sometimes occurs in the striped grey area depending on the value of ρ_{AC}.

block-structured matrix remains unchanged when considering the stability of B and C partially fragmented states. It is only the parameters and the elements of the blocks that comprise the block-structured matrix that change when considering B and C partially fragmented states.

Due to the asymmetry in the S.T.V. model we had to repeat the whole procedure to produce phase diagrams for the B partially fragmented state. Fortunately, due to the symmetry between dynamics of the A and C nodes we could produce the bifurcation plots for the C partially fragmented state with little extra work. When considering the stability of the B partially fragmented state we can use the same moment closure approach as employed by Bohme and Gross [19] to provide some bounds on when B partial fragmentation occurs. In particular, we investigate two regions: $\rho_{AC} = 0$ and $\rho_{AC} = \rho_{ACmax}$. The value for ρ_{ACmax}

$$\rho_{ACmax} = \frac{k-1}{2k}, \tag{47}$$

is obtained through the moment closure method. We can identify two distinct regions in the B partial fragmentation phase diagram, Figure 11. In the white shaded area, B partial fragmentation never occurs and the final state of the system depends on p_2. In the grey striped shaded area, B partial fragmentation occurs depending on the value of ρ_{AC}. Specifically, we find that B partial fragmentation never occurs when $\rho_{AC} = \rho_{ACmax}$ and that it always occurs when $\rho_{AC} = 0$, $p_1 > p_f^1$ and $p_3 > p_f^1$. We find that it sometimes occurs for $0 < \rho_{AC} < \rho_{ACmax}$, $p_1 > p_f^1$ and $p_3 > p_f^1$. In order to predict the final state of the full system we must consider the partial fragmentation phase diagrams for each of the three states. We assume that

$$p_1 \geq p_2 \geq p_3. \tag{48}$$

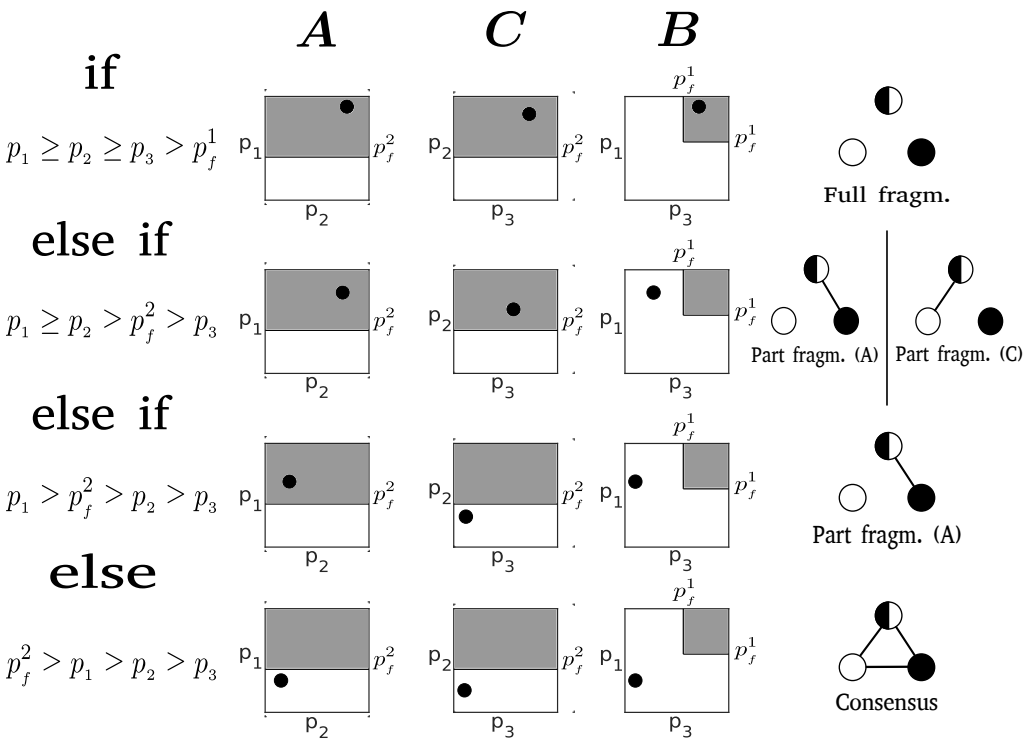


Figure 12: Phase diagrams for partial fragmentation with respect to each state A, B and C for $p_1 \geq p_2 \geq p_3$ in the S.T.V. model. The grey shaded area in each diagram illustrates when partial fragmentation with respect to that state occurs. The positions of the black circles within each diagram indicate whether or not partial fragmentation with respect to that state occurs or not. The final state of the system is shown on the right-hand side of the phase diagrams.

Given (48) and that $p_f^1 > p_f^2$, there are four distinct cases to analyse:

$$\begin{aligned}
&\text{Case} \quad 1: p_1 \geq p_2 \geq p_3 > p_f^1, \\
&\text{Case} \quad 2: p_1 \geq p_2 > p_f^2 > p_3, \\
&\text{Case} \quad 3: p_1 > p_f^2 > p_2 > p_3, \\
&\text{Case} \quad 4: p_f^2 > p_1 > p_2 > p_3.
\end{aligned}$$

In Case 1, full fragmentation occurs reached because partial fragmentation occurs for each state A, B and C. In Case 2, partial fragmentation occurs for states A and C and not for state B. This means that the system will either reach: an A partially fragmented state where the active BC component will eventually reach internal consensus on state C or a C partially fragmented state where the active AB component will eventually reach internal consensus on state A. In this case, the specific values of p_1,p_2 and p_3 will determine which of these partially fragmented states is reached. In Case 3, A partial fragmentation occurs and the active BC component will eventually reach internal consensus on state C. In Case 4, no partial fragmentation occurs and therefore the system reaches a global consensus state where all nodes have the same belief state. The consensus belief state will either be A or C and this will depend on the specific values of p_1, p_2 and p_3.

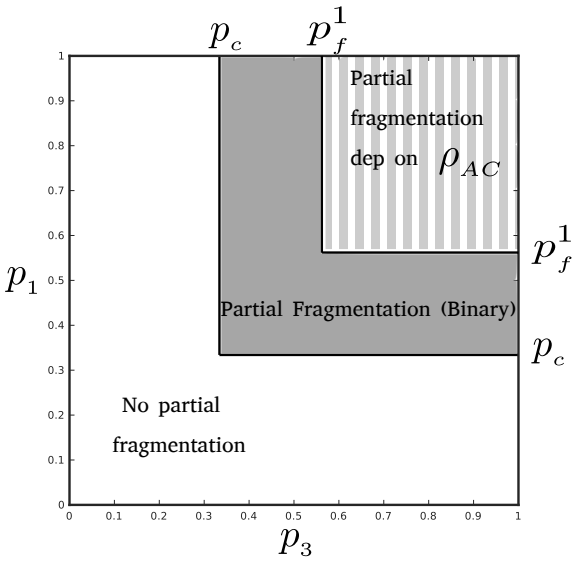


Figure 13: Comparison of the phase diagrams for B partial fragmentation in the simplified B.V. and T.V. models.

In summary, we have shown that full fragmentation only occurs if the rewiring rates, p_1, p_2 and p_3 exceed the critical rate p_f^1. Through a similar analysis Bohme and Gross [19] conclude that full fragmentation only occurs in the S.B.V. model if the rewiring rates p_1, p_2 and p_3 exceed the critical rate p_c. Figure 13 shows that the critical rate required for full fragmentation to occur is greater in the S.T.V. model than in the S.B.V. model explored by Bohme and Gross [19],

$$p_f^1 > p_c. \tag{49}$$

This demonstrates how the inclusion of explicit uncertainty in the adaptive voter model, in the form of the introduction of an uncertain belief state, increases the critical rewiring rate required to transition the system to a fully fragmented state. From this we can conclude that the inclusion of uncertainty makes the adaptive voter model more robust in the idealised case, as it delays full fragmentation of the network when there is no noise and there are no zealots.

3.4.1 Random state change development

We can investigate how the presence of noise affects the fragmentation transition for both the S.B.V. and S.T.V. models by including the random state change event in the derivation. We extend the S.B.V. model as presented by Bohme and Gross [19] and the S.T.V. model presented in Section 3.1.1. We follow the some procedure that we used to identify the fragmentation transition for the S.T.V. model in Section 3.4

We consider partial fragmentation with respect to A nodes. We derive new transition rules for the 1 and 2-fans depicted in Figure 4 that include the random state change event. We provide the complete picture and a full description for the transition rules governing the type-1 q-fan in the S.T.V. model and choose to omit an explicit description of the transition rules for the rest of the q-fans from both models.

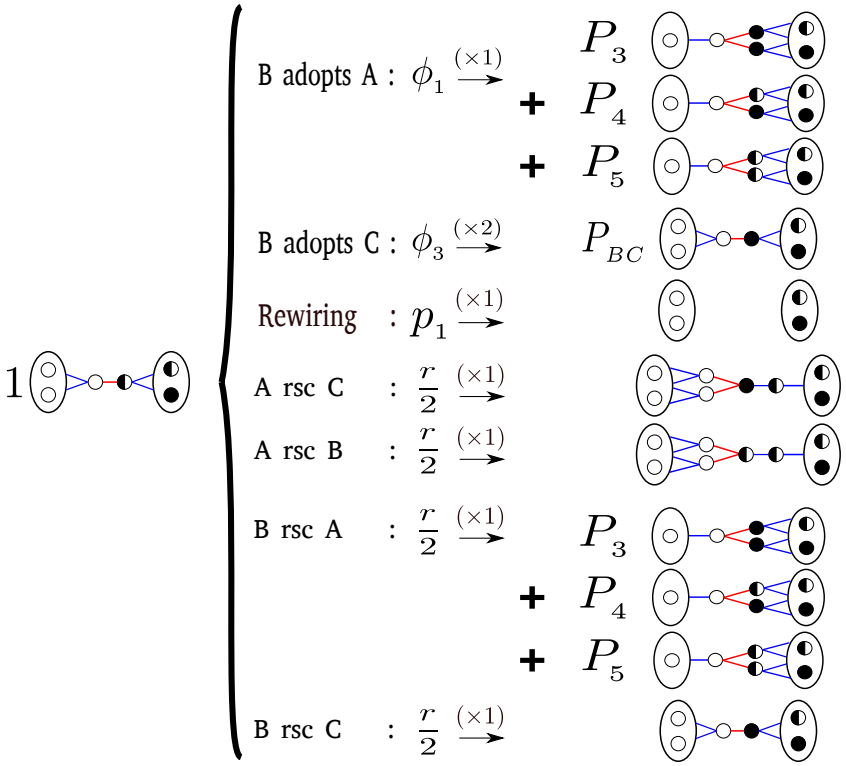


Figure 14: Complete picture of the transition rules governing the type-1 q-fan in the three-state model with uncertainty. Active links are coloured red and inert links are coloured blue.

There are 4 random state change events that can occur on this type-1 q-fan: the A node randomly changes state to a B or C node and the B node randomly changes state to an A or C node. All of these events occur with the same probability $\frac{r}{2}$ and a type-1 q-fan is destroyed when each event occurs. When the A node randomly changes state to a C node a type-6 q-fan is created as the C node is now connected to the 2 A nodes that the A node was previously connected to. When the A node randomly changes state to a B node a type-7 q-fan is created as the B node is now connected to the 2 A nodes that the A node was previously connected to. When the B node randomly changes state to an A node a type-3 q-fan is created with probability P_3 if the B was connected to two C nodes, a type-4 q-fan is created with probability P_4 if the B was connected to one B node and one C node and a type-5 q-fan is created with probability P_5 if the B node was connected to two B nodes. When the B node randomly changes state to a C node a type-2 q-fan is created. The derived transition rules define a dynamical system that describes how the density of 1 and 2-fans evolve over time.

3.4.2 Zealot development

We can also investigate how the presence of zealots affects the fragmentation transition for both the S.B.V. and S.T.V. models. We extend the S.B.V. model as presented by Bohme and Gross [19] and the S.T.V. model. We follow the some procedure that we used to identify the fragmentation transition for the S.T.V. model in Section 3.4

As stated in Chapter 1, a zealot is an agent that does not change its opinion. More specifically, there is no change to the network if a zealot is picked to adopt the belief state of another agent in an opinion adoption event. In addition, zealots are unaffected by random state change events.

Given a finite density of zealots in the network, ρ_z, we can deduce that the probabilities that a given node is and is not a zealot are

$$\Pr(\text{Node is a zealot}) = \rho_z \quad \text{and}$$
$$\Pr(\text{Node is not a zealot}) = \left(1-\rho_z\right).$$

We consider partial fragmentation with respect to A nodes. We derive new transition rules for the 1 and 2-fans depicted in Figure 4 that account for the presence of zealots. In short, the presence of zealots in the network can be captured by appending $\left(1-\rho_z\right)$ as a prefactor to each term that describes the probability for a given opinion adoption event and a given rewiring event to occur. As a concrete example consider the opinion adoption event that occurs on a type-1 q-fan when the B node adopts the opinion of the A node. This event occurs with probability ϕ_1 in the original model, when there are no zealots. When there are zealots in the network this event only occurs if the B node is not a zealot. This event therefore occurs with probability $\left(1-\rho_z\right)\phi_1$. The derived transition rules define a dynamical system that describes how the density of 1 and 2-fans evolve over time.

3.4.3 Results

In order to investigate the effect of presence of noise, in the form of the random state change event, and the effect of individual failure, in the form of finite zealot density, on the fragmentation transition we make the simplification $p_1 = p_2 = p_3 = p$. With this simplification we can determine the rewiring rate p required to transition the S.B.V. and S.T.V. models to a fully fragmented state. We generated the following plots by evaluating $\lambda(\boldsymbol{J})$ for each state along the one dimensional axis $p \in [0,1]$ for each value of the relevant variable, r or ϕ_z, in the range $[0,1]$. Following the same method used in Section 3.4 we claim that full fragmentation occurs when partial fragmentation with respect to each state occurs.

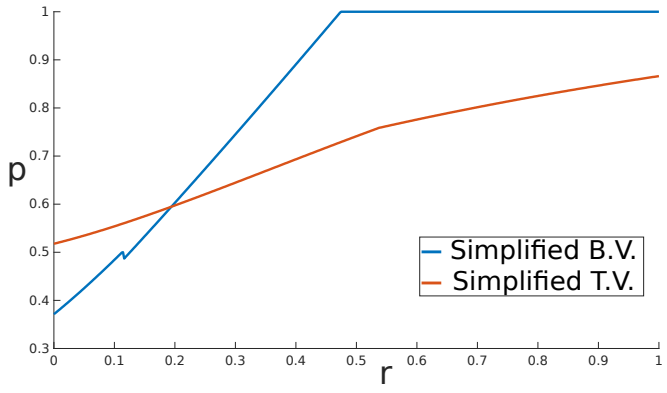


Figure 15: Fragmentation transition as a function of random state change event rate r for the S.B.V. and S.T.V. models with $p_1 = p_2 = p_3$

Figure 15 depicts the rewiring rates required to transition the S.B.V. and S.T.V. models to a fully fragmented state for different values of the random state change event rate r. It can be seen that, for small values of r the S.T.V. model requires a higher rewiring rate p to transition

to a fully fragmented state whereas the opposite is true for large values of r. From this, we can conclude that the S.T.V. model has a higher noise tolerance than the S.B.V. model for small levels of noise as it exhibits a greater efficacy in delaying full fragmentation of the network and vice versa for large levels of noise.

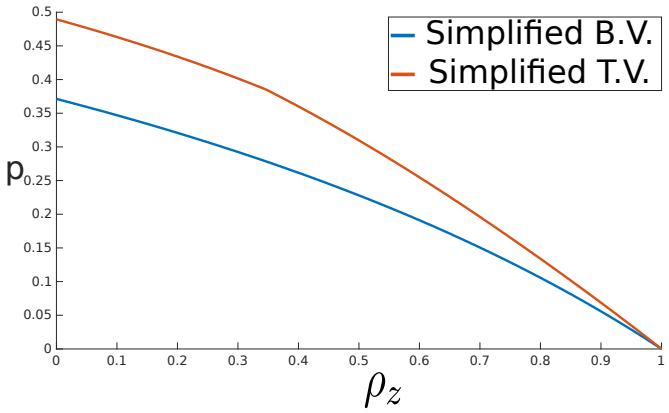


Figure 16: Fragmentation transition as a function of zealot density ρ_z for the S.B.V. and S.T.V. models with $p_1 = p_2 = p_3$

Figure 16 depicts the rewiring rates required to transition the S.B.V. and S.T.V. models to a fully fragmented state for different values of the zealot density ρ_z. It is clear that for all values of ρ_z the S.T.V. model requires a higher rewiring rate p to transition to a fully fragmented state. This means that the S.T.V. model has a higher fault tolerance than the S.B.V. model as it exhibits a greater efficacy in delaying full fragmentation of the network.

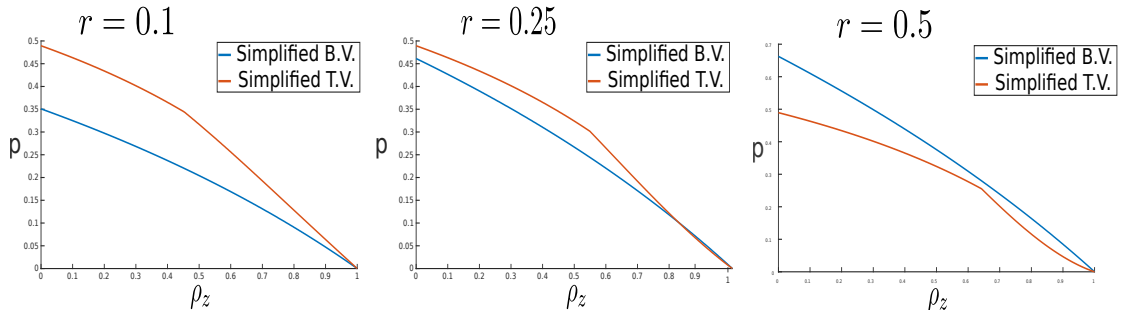


Figure 17: Fragmentation transition as a function of zealot density ρ_z for particular values of the random state change event rate r for the S.B.V. and S.T.V. models with $p_1 = p_2 = p_3$.

Figure 17 illustrates the combined effect of the presence of noise and zealots on the fragmentation transition in the S.B.V. and S.T.V. models. It shows that for small values of noise r, the S.T.V. model exhibits a more delayed onset of full fragmentation than the S.B.V. model for all values of ρ_z and vice versa for large values of r. This demonstrates that the S.T.V. model has a higher fault tolerance than the S.B.V. model for low noise levels and vice versa.

4 Agent-based simulation

This Section presents the results obtained from stochastic agent-based simulations of the complete T.V and B.V models described in Section 2. Simulation of stochastic dynamics on

complex networks is generally a computationally expensive task. We choose to extend an exact stochastic simulation algorithm known as the Doob-Gillespie algorithm [33] over traditional Monte-Carlo methods. This is because traditional Monte-Carlo methods use rejection-sampling techniques which involve checking whether each possible event has occurred at every time-step [34]. On the contrary, the Doob-Gillespie algorithm samples the time taken until the next event occurs and which type of event occurs at this time [34]. This significantly reduces the time required to simulate a stochastic process on a given network. Vestergaard and Génoiset [34] show that it can be up to 500 times faster than a classical rejection sampling algorithm. It was not trivial to adapt the standard Doob-Gillespie algorithm to the T.V and B.V models due to the dynamic nature of the network topology and the diversity of events that can occur.

The task of simulating the models for various parameter ranges is an Embarrassingly Parallel computational problem. This means that due to the independence between simulations for different parameter values it did not require much effort to run parameter sweeps in parallel using Bristols high performance computing machine Blue Crystal.

4.1 Results

Our simulation architecture captures the complete T.V. and B.V. models. It can be used to explore the role of the following

- Number of agents in the network - N,
- Inconsistency threshold - I,
- Rewiring rate - p,
- Random state change event rate - r,
- Zealot density - ρ_z,
- Opinion adoption rate - ϕ,
- Number of propositions - n.

We choose to focus on just varying r and ρ_z in order to explore the fault tolerance and noise tolerance of the models. The benefit of agent-based simulations is that we have more freedom when defining which metrics to use in order to measure the fault tolerance and noise tolerance of the models.

We choose to measure the fault tolerance of the models with the following metrics:

- Metric 1: Average size of a zealots component,
- Metric 2: Average payoff of nodes.

We claim that Metric 1 provides an evaluation of the magnitude of a zealots influence. The smaller the average size of a zealots component the less influence the zealot has on the rest of the nodes in the network. We proceed to suggest that zealots will have a smaller influence in a model with a higher fault tolerance. In addition, Metric 2 provides a measure of success of the algorithm in solving the best-of-n problem. A model with a higher fault tolerance will produce a

higher average payoff for nodes in the network. To clarify, a lower score for Metric 1 implies a higher fault tolerance and a higher score for Metric 2 implies a higher fault tolerance.

We choose to measure the noise tolerance of the models with the following metrics:

- Metric 3: Probability of reaching consensus,
- Metric 4: Average payoff of nodes.

We do not consider Metric 1 for obvious reasons and we do not consider Metric 3 when measuring the fault tolerance of the models due to the reason outlined in Section 3.3.3, namely that consensus on a zealots belief state is a suboptimal solution. We claim that Metric 3 is a good measure for the success of the algorithm as the best-of-n problem requires a consensus. The reasons for choosing Metric 4 are the same as for Metric 2. To clarify, a higher score for both Metrics 3 and 4 implies a higher noise tolerance.

4.1.1 Fault tolerance

Figures 18 and 19 were generated by measuring the long-time limit statistics of the network, averaging over 20 realisations, with the following parameters:

N	I	p	r	ρ_z	ϕ	n
100	1	0.001	0	ρ_z	0.999	5

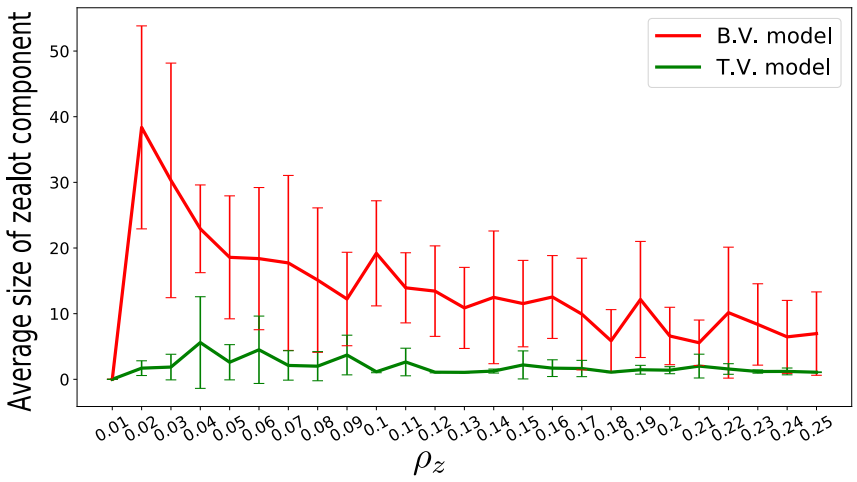


Figure 18: Metric 1 as a function of ρ_z for both the T.V. and the B.V. models.

Figure 18 shows that the T.V. model scores lower on Metric 1 than the B.V. model. This demonstrates that the T.V. model has a higher fault tolerance when measured by this metric.

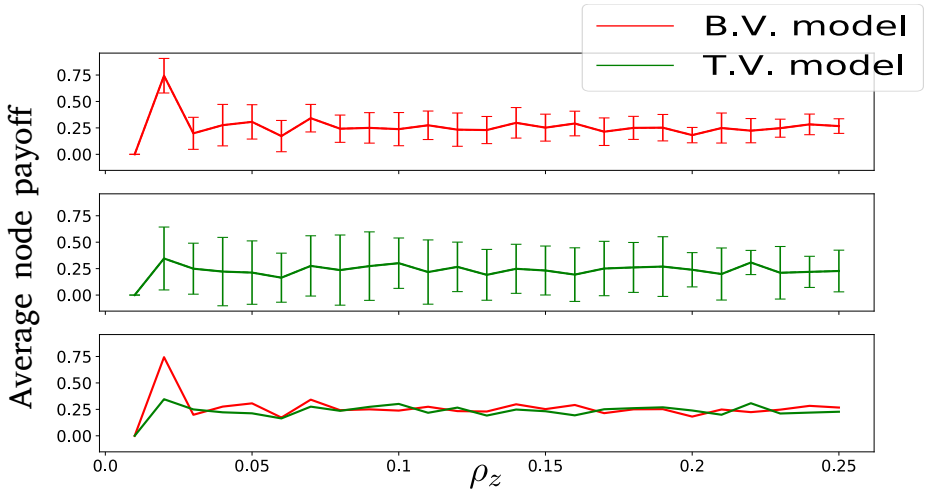


Figure 19: Metric 2 as a function of ρ_z for both the T.V. and the B.V. models.

Figure 19 shows that for small zealot density ρ_z the B.V. model scores higher on Metric 2 than the T.V. model whilst for larger zealot densities $\rho_z > 0.05$ both models exhibit similar scores for Metric 2. This shows that the B.V. model has a higher fault tolerance when there is a small number of individual failures in the system.

4.1.2 Noise tolerance

Figures 20 and 21 were generated by measuring the long-time statistics, averaging over 20 realisations, with the following parameters:

N	I	p	r	ρ_z	ϕ	n
100	1	0.001	r	0	$(1-r)$	5

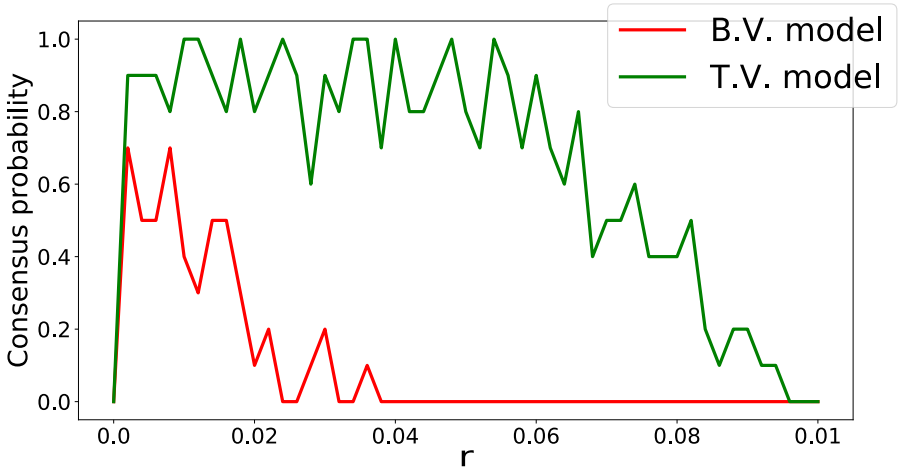


Figure 20: Metric 1 as a function of r for both the T.V. and the B.V. models.

Figure 20 shows that the T.V. model scores higher on Metric 3 than the B.V. model for small amounts of noise $0 < r < 0.01$. It is also shows how both models score poorly for $r > 0.01$. We suggest that this means the T.V. model has a higher noise tolerance for low levels of noise but both models have a low noise tolerance for high levels of noise.

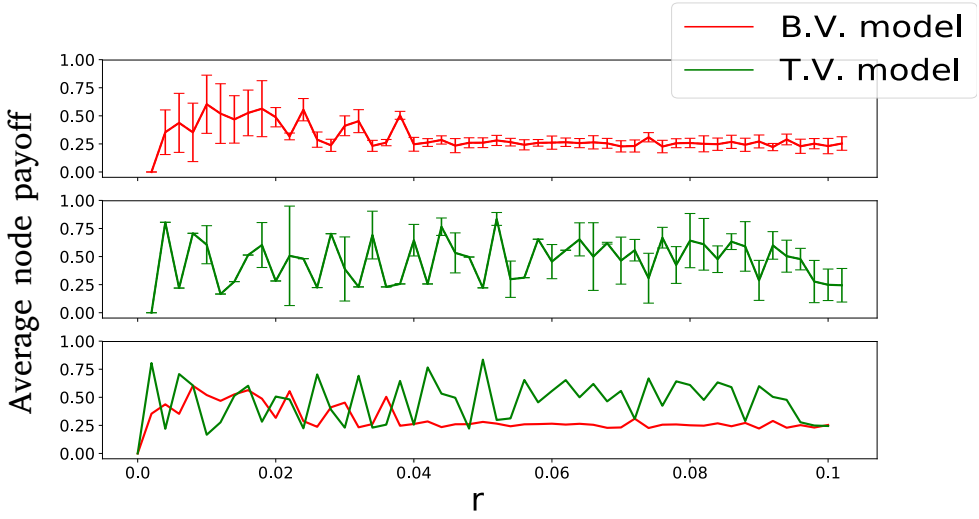


Figure 21: Metric 4 as a function of r for both the T.V. and the B.V. models.

It is hard to make any concluding remarks about the difference between the two models in Figure 21 for $r < 0.04$. For larger values of r we can say that the T.V. model scores higher, on average, for Metric 4 than the B.V. model. This means that the T.V. model has a higher noise tolerance than the B.V. model for relatively low levels of noise.

5 Conclusion

5.1 Project review

This report has investigated how the addition of uncertainty affects the fault and noise tolerance of a decentralised decision-making algorithm known as the adaptive voter model. We began by presenting a model for an extended n state adaptive voter model, which we call the T.V. model, that includes weighted propositions, some notion of inconsistency between belief states, uncertain belief states, a random state change event and the presence of zealots. In addition to this extended model, we also derived an equivalent model, which we call the B.V. model, that does not include uncertain belief states. We proceeded by deriving simplified versions of the extended models, referred to as the S.T.V. model and the S.B.V. model, with the aim of performing some analysis. Finally, we performed agent-based simulations with both the T.V. and the B.V. models.

In particular, we analytically demonstrated that the S.T.V. model takes less time to converge on a consensus state than the B.T.V. model. In addition, we showed that the S.T.V. model has a higher noise tolerance than the B.T.V. model for low noise levels. We also analytically demonstrated that the S.T.V. model exhibited a greater resistance to transitioning to a fully fragmented state than the B.T.V. model. Furthermore, we found that the S.T.V. model has a higher fault tolerance for small noise levels and a smaller fault tolerance for large noise levels. These analytical results were supported by the findings from the agent-based simulations. We also believe that the work has shed some light on the role of uncertainty in preventing polarisation and promoting consensus in adaptive social networks.

5.2 Further work

The analytical method for computing the consensus time in the S.T.V. and B.T.V. models could be extended by employing the approximation presented by Di Patti et al., [30] to obtain an analytical expression for the scaling of the consensus time with the network size N. Failing this, a more thorough numerical investigation could be carried out using our numerical implementation of Gillespie's algorithm [31]. The analytical method for computing the fragmentation transition in the S.T.V. and B.T.V. models could be extended by considering the spider motif base as opposed to the q-fan motif base as this motif base more accurately captures the degree distribution of the network [18].

Our simulation architecture for the agent-based simulations of the full T.V. and B.V. models could be improved to increase efficiency and run-time performance. This would allow a more extensive study on these models for larger network sizes. In addition, more agent-based simulations of the S.T.V. and the B.T.V. should be carried out in order to check the accuracy of the analytic results presented.

Finally, it would be great to test the accuracy of the analysis and simulation results with an embodied experiment in a real swarm robotic system. This would demonstrate whether or not the extended model provides a solution to the problem of fault and noise tolerance in decentralised decision-making algorithms in robotic swarms.

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